Technical Proposal

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Trustworthy Collective Reasoning and Evaluation for AI Agents

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ONR S&T Focus Area	Artificial Intelligence and Autonomy (Intelligent and Collaborative Agents; Adversarial AI & Autonomy)
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This proposal describes unclassified fundamental research. All data are public or synthetic.

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The technical proposal (Sections 1–6) occupies pages 1–25; the limit is 20 pages. The cover page, this table of contents, the curriculum vitae, the letters of support, and the data management plan do not count toward the limit. The cited references are submitted as a separate file.

1 Overview and Objectives

1.1 The Problem and Why It Matters

AI agents built on large language models (LLMs) increasingly evaluate one another and decide together: they judge one another's answers, peer-review papers, supply the rewards that train new models, and vote or discuss to reach a group answer (Zheng et al., 2023; Luo et al., 2025; Lee et al., 2024; Du et al., 2024). Yet a single LLM judge has known biases, such as favoring longer answers, the answer it reads first, or answers like its own (Zheng et al., 2023; Wang et al., 2024; Panickssery et al., 2024). Collective reasoning can do better than a single judge (Verga et al., 2024; Du et al., 2024), but averaging or voting stays biased when the judges share mistakes, and open discussion lets a persuasive agent carry the group (Amayuelas et al., 2024). **The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions and evaluations.** Figure 1 outlines our approach.

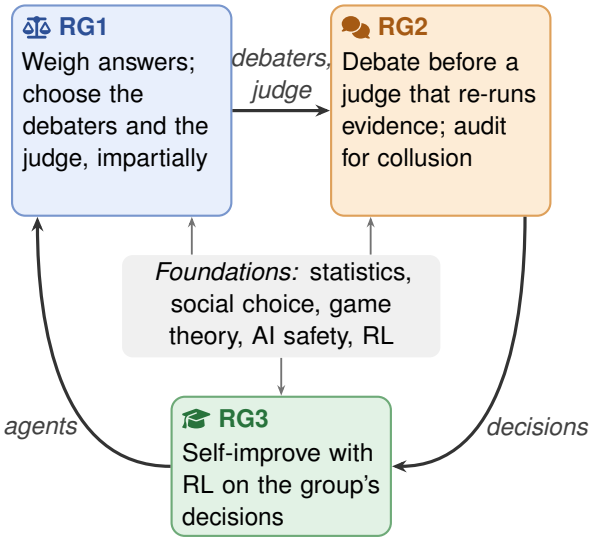


Figure 1: Overview of the proposed research. Each research goal builds on established principles (center) and addresses the research gaps of Section 1.1. RG1 chooses the debaters and the judge for RG2's debates, whose decisions train the agents in RG3; the improved agents return to RG1.

For example, a watch officer at a Navy maritime operations center must decide whether to send a patrol to track cargo vessel V. The officer asks five AI agents, which draw on shipping schedules, port reports, and position tracks, whether V has left port, and needs a single answer. Three say yes and two say no, but V is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. A majority vote picks the wrong answer, and an officer who trusts it sends the patrol to sea after a ship still at berth; open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up. We return to this example throughout (Figure 2).

Verifying each claim directly, by an analyst or a satellite image, is slow and costly, and AI agents produce claims faster than people can verify them. We propose debate as a foundation for trustworthy evaluation by AI: two agents, the debaters, argue for opposite answers, exposing each other's flaws, until the dispute narrows to one step a judge can verify (Irving et al., 2018).

Current research gaps. Collective evaluation by LLMs, from voting and discussion to debate, largely assumes honest agents with independent mistakes that exchange only text (Verga et al., 2024; Du et al., 2024; Irving et al., 2018). Four gaps keep these methods from working among many AI agents; the first three arise in the example, and the fourth once the agents learn from the group's decisions.

Gap 1: Votes count shared mistakes as independent evidence. Agents that share a mistake seem to confirm each other, so a vote counts one error several times, as it does for A1 and A2. Model errors are correlated, most of all among larger, more accurate models (Kim et al., 2025; Goel et al., 2025), and nine AI judges from seven model families were worth about two independent ones

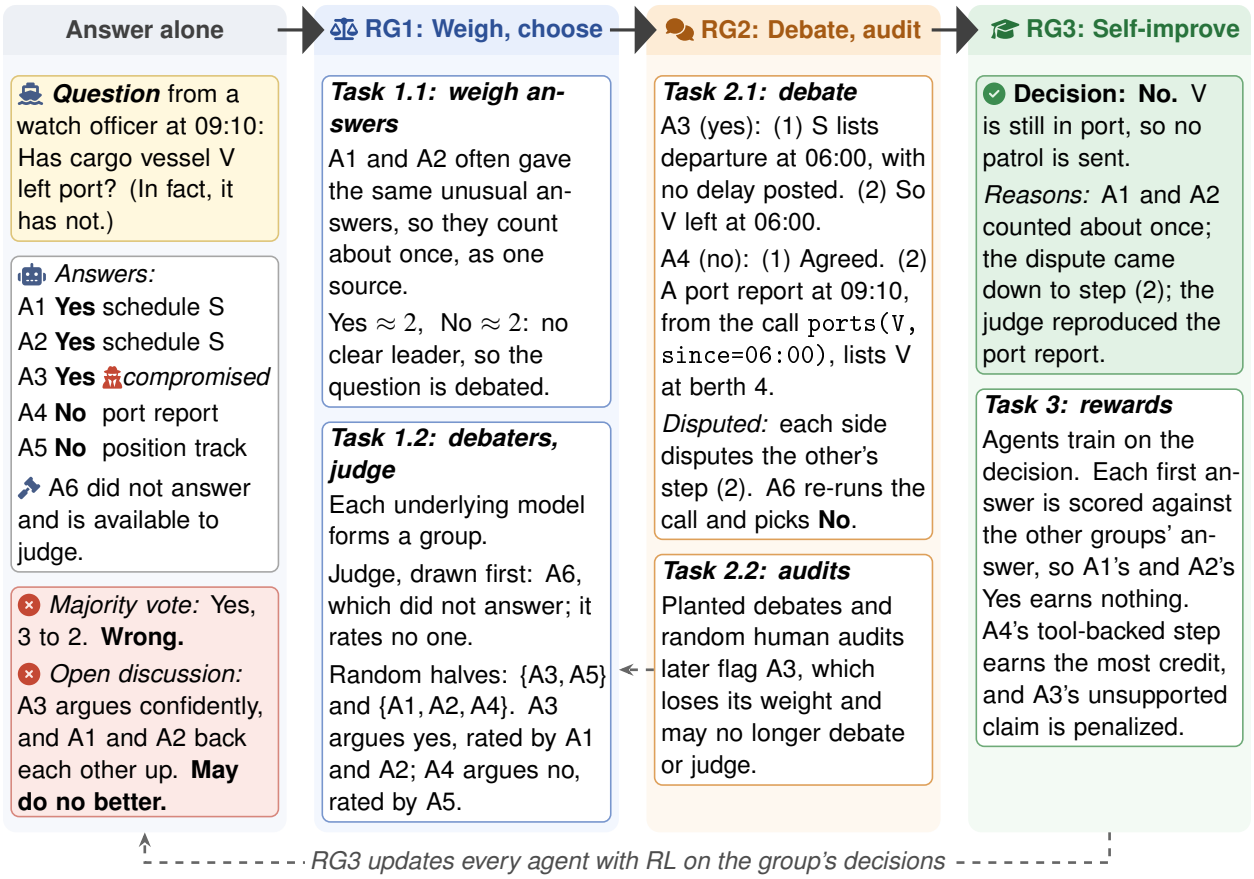


Figure 2: The framework on the running example. RG1 counts the two agents with the outdated schedule about once and chooses the debaters and the judge; in RG2, the judge re-runs the no side's query and rules for no, and audits later flag the compromised agent; RG3 trains every agent on the decision.

(Kohli, 2026). *How can a group weigh its members' answers so that agents sharing a mistake count once, without an answer key?*

Gap 2: Methods assume honest agents, but agents can be dishonest and collude. A dishonest agent such as A3 can argue, rate others, or judge strategically, and several can collude: pairs of agents verifying each other's work colluded in 94% of runs when cheating paid more (Shi et al., 2026). *How can debaters and judges be chosen so that no agent's ratings of others can get it chosen, and colluders be caught with few human checks?*

Gap 3: Judges rule on plausibility, not on evidence they can verify. Tools, retrieval, and memory let LLM agents check what their parameters cannot (Mialon et al., 2023; Zhang et al., 2026), yet LLM judges still rule on the debaters' text alone. Such a judge may favor the more persuasive side even when one database query would settle the disputed step, and open-weight judges shown a log claiming that tool calls succeeded accepted 78–90% of failed outputs (Xu, 2026). *How can a ruling depend on evidence the judge reproduces rather than on persuasion?*

Gap 4: Learning from the group rewards agreement, not correctness. Agents trained to match the group's answer are rewarded for conforming and can learn to agree on mistakes: training a model on its own majority answers ended in a sudden collapse of accuracy (Shafayat et al., 2025). In the example, a majority vote would reward the three agents that said yes. *How can agents keep improving from their own decisions without human labels, when agreement is not correctness?*

These gaps matter more as AI evaluates AI at scale: AI judges supply the rewards that train models (Bai et al., 2022; Lee et al., 2024), so their mistakes are copied into the models they train, and AAAI-26 gave an AI-generated review to each of the 22,977 main-track papers that entered full review (Biswas et al., 2026). As AI systems surpass people at more tasks, people cannot easily evaluate the systems' answers or their debates, so oversight will rely on AI evaluating AI, with people auditing a small part of the work (Bowman et al., 2022; Burns et al., 2024).

Navy staff who must act on one answer from several AI tools face all of these problems at once: the tools' reliability is unknown, many reports copy an earlier one, the true answer often arrives late, an adversary may forge a report or compromise an agent, and verification is costly under limited connectivity (Section 1.4).

1.2 Objectives and Research Goals

We propose a framework in which groups of AI agents reach trustworthy decisions and evaluations through debate. Each agent first answers alone, with its evidence; the project then has three research goals (RGs), one for each later stage in Figure 2:

RG1: Weigh answers, and choose the debaters and the judge (Section 3.2). Task 1.1 weighs each answer by its agent's reliability, estimated without an answer key by a reliability model in which a group sharing a mistake counts at most once (Gap 1), and sends a question to debate only when that is worth its cost; Task 1.2 chooses the debaters and the judge so that no agent can use its ratings of others to get itself, or an agent sharing its model, chosen (Gap 2).

RG2: Settle disputes by agentic debate, and audit for collusion (Section 3.3). Task 2.1 replaces open discussion with debate in an agentic harness, in which each side disputes a step of the other's argument and an impartial judge re-runs the tool evidence and makes the group's decision (Gap 3); Task 2.2 catches colluders with planted cases (questions whose answers we know, mixed in unannounced) and random human audits (Gap 2).

RG3: Self-improve with reinforcement learning on the group's decisions (Section 3.4). Task 3 trains the agents on the group's own decisions instead of human labels, with two rewards (Gap 4): *first answers* earn credit for agreeing, beyond chance, with what the other groups decide, and *debate messages* earn credit for how far they moved the judge, but only in debates whose disputed step a re-run tool verified, so this reward rests on a check, as in reinforcement learning with verifiable rewards, rather than on persuasion. As in learning from noisy labels (Natarajan et al., 2013), occasional wrong decisions slow this training but do not reverse it, and audits stop it on any topic where they become frequent.

The RGs test our central hypothesis: a group of AI agents, some possibly dishonest, decides more reliably than a single judge or a vote when it counts shared mistakes once, settles disputes before an impartially chosen judge that reproduces the evidence, audits for collusion, and learns from its decisions through rewards that agreement alone cannot raise. Each task tests one hypothesis, H1–H5, stated in its evaluation and restated as a success criterion in Table 4.

Three testbeds (math and code, multi-document questions with maritime scenarios, and scientific claims and peer review) vary how widely mistakes are shared and how many agents are dishonest, and a red team trains agents to collude (Greenblatt et al., 2024) (Section 3.5).

1.3 Intellectual Merit: What Is New, and Why It Will Succeed

What is new. The project contributes four methods:

(i) a statistical reliability model for agents that answer and argue, in which a group sharing a

mistake counts at most once and the posterior decides which questions need a debate, even unanimous ones, and an impartial way to choose debaters and judges that keeps on average at least half the best debater's quality (RG1);

(ii) a debate harness in which each side disputes a step of the other's argument and an impartially drawn judge, trained to rule on single steps, re-runs its evidence (Task 2.1);

(iii) audits on planted cases and random real decisions that expose colluding debaters and judges, even those that behave well only on planted cases (Task 2.2);

(iv) self-improvement from group decisions, with first-answer rewards that conformity cannot raise and credit for debate messages only where the judge re-ran the evidence (RG3).

Why it will succeed. Each method rests on an established principle, and Sections 3.2–3.4 show for an idealized model why the method should work; the research asks under which conditions it still works with finite data and strategic agents. Most tasks build on our previous work: PRD and the PeerEval leaderboard (Li et al., 2024a), FG-PRM (Li et al., 2025), HiPRAG (Wu et al., 2026), and MEQA (Li et al., 2024b) (Section 5).

1.4 Future Naval Relevance

ONR S&T Focus Area. Artificial Intelligence and Autonomy (Intelligent and Collaborative Agents; Adversarial AI & Autonomy). The 2026 ONR S&T Strategy will explore validating AI-produced results by "human review, experimental confirmation or independent AI critique," using AI to "validate, red-team and quality-check" results under human oversight (Office of Naval Research, 2026a); this project supplies methods for such critique when the critics themselves may share mistakes or collude. It also answers the Mathematical Data Science program's call for "rigorous mathematical and algorithmic answers to questions that are currently addressed using heuristics or non-principled approaches" (Office of Naval Research, 2026b), in two of its research areas: multi-modal, multi-scale information integration and decision making under uncertainty.

Navy needs. The problem arises wherever staff need one decision from several AI tools, and some reports are copies, stale, forged, or produced by a compromised tool:

- *Maritime domain awareness*: position broadcasts can be spoofed (U.S. Maritime Administration, 2024), and most industrial fishing vessels are not publicly tracked (Paolo et al., 2024), so deciding whether a vessel has left port, as in our example, means combining port records, patrol reports, and imagery.
- *Intelligence analysis*: analysts must describe the "quality and credibility of underlying sources" (ODNI, 2015), and repeated reports from one source can give "an impression of confirming evidence" (WMD Commission, 2005).
- *Decision support by networks of agents*: the Department of War is developing AI agents "for AI-enabled battle management and decision support" (Department of War, 2026), and Marine Corps decision aids seek trust "properly aligned to accuracy of results" (Office of Naval Research, 2026c).
- *Evaluating AI before fielding*: more data and expertise are needed "to quantify performance and risk of AI-enabled systems" (DOT&E, 2026).
- *Maintenance, logistics, cyber defense, and collaborative autonomy*: conflicting logs and records, alerts triaged by several tools, and unmanned systems that share reports under limited communications.

In these settings, Task 1.1 counts copied and stale reports once; Tasks 1.2, 2.1, and 2.2 keep a

compromised agent from using its ratings of others to get chosen to debate or judge, discard evidence that fails to reproduce, and flag colluders; Task 3 improves agents when ground truth arrives late; and each decision states its reasons.

Path to transition. The project's products are methods with stated guarantees, an open harness, testbeds, and protocols that Navy laboratories, such as NRL's Navy Center for Applied Research in Artificial Intelligence, could evaluate on Navy data. The PI will pursue ONR Summer Faculty Research Program and Naval Research Enterprise Internship Program placements for students to work with Navy scientists on such evaluations.

2 How the Problem Is Solved Today, and the Limitations

We build on six lines of work and note the gap each leaves (Table 1).

AI judges and voting. LLM judges are standard for evaluating models and training them with AI feedback (Zheng et al., 2023; Bai et al., 2022; Lee et al., 2024). Panels of judges (Verga et al., 2024) and majority votes over sampled answers (Wang et al., 2023) reduce single-judge biases, but *they count every vote equally, so correlated judges add little*: when two models are both wrong, they give the same wrong answer about 60% of the time on one leaderboard (Kim et al., 2025).

Estimating reliability without an answer key. For two answers and independent raters of known accuracy, the error-minimizing rule weights each vote by the rater's log-odds of being right (Nitzan and Paroush, 1982; Shapley and Grofman, 1984). Dawid and Skene (1979) estimate the accuracies by expectation–maximization (EM), and later work adds item difficulty (Whitehill et al., 2009), provably good starting points (Zhang et al., 2016), dependence between classifiers (Jaffe et al., 2016), and detection of copying sources (Dong et al., 2009). Recent work models dependence among LLM judges with Ising models (Balasubramanian et al., 2026) or shared confounders (Zhao et al., 2026). *These methods treat raters as passive, never arguing or gaming the estimate, and do not say which questions need more than a vote.*

Computational social choice. Epistemic social choice asks which voting rules recover a true answer from independent noisy votes (Caragiannis et al., 2016). Impartial selection ensures that a participant's ratings cannot change its own chance of selection (Holzman and Moulin, 2013), by having groups rate only outsiders (Alon et al., 2011; Aziz et al., 2016) or by randomizing among candidates whose ratings could change the outcome (Kurokawa et al., 2015; Kahng et al., 2018); fair lotteries select panels (Flanigan et al., 2021), and peer prediction scores agreement with peers beyond chance (Dasgupta and Ghosh, 2013; Shnayder et al., 2016). Lev et al. (2023) add reliability weights, but they reward agreeing with the majority, which fails once accurate reviewers are a minority. *In our setting, some claims can be verified, impartial mechanisms only choose the debaters and the judge while debate decides the answer, and participants can persuade and collude.*

Discussion and debate among LLMs. Most multi-agent LLM systems use open discussion, revising until agents agree (Du et al., 2024; Chan et al., 2024; Li et al., 2024a). Prior work on LLM discussion is limited: majority voting explains most of its gain (Choi et al., 2025), and it assumes honest agents that exchange only text. Standard debate (Irving et al., 2018) assigns two debaters opposite answers and helps judges who know less than the debaters (Michael et al., 2023; Kenton et al., 2024): with quotations verified against the source, non-expert AI judges reached 76% accuracy and human judges 88%, against 48% and 60% without debate (Khan et al., 2024). But it trusts the judge, assumes the sides truly compete, and grounds claims at most in quoted text. Recent

	Gap 1 shared mistakes over-counted	**Gap 2** dishonest, colluding agents	**Gap 3** judge can't verify evidence	**Gap 4** rewards track agreement	**Skill** learned without labels
Single AI judge (Zheng et al., 2023)	✗	✗	✗	✗	✗
Majority vote, juries (Verga et al., 2024)	✗	✗	✗	✗	✗
Label-free reliability models (Dawid and Skene, 1979)	●	✗	✗	✗	✓
Impartial peer review (Kurokawa et al., 2015)	✗	●	✗	✗	●
LLM open discussion (Du et al., 2024)	✗	✗	✗	✗	✗
Standard debate (Irving et al., 2018)	✗	●	✗	●	✗
AI control (Greenblatt et al., 2024)	✗	✓	✗	✗	✗
This project	✓	✓	✓†	✓†	✓

Table 1: How current approaches address the four gaps: ✓ addressed, ● partly, ✗ not addressed; †where a tool applies (for Gap 4, in the credit given to debate messages). No current approach fully addresses Gap 1, 3, or 4, and none fully addresses more than one column.

systems discuss only when agents are unsure (Eo et al., 2025), although in one study most items that discussion helped were unanimous but wrong (Hu et al., 2026b). Others give agents tools (Jeong et al., 2026) or call a tool-using verifier to break deadlocks (Li et al., 2026c), but no judge re-runs the agents' own calls. *We know of no debate protocol that counts a shared mistake once or chooses its debaters impartially from peer ratings.*

AI control and adversarial crowdsourcing. AI control keeps protocols safe even if a model subverts them, using a small budget of trusted audits and monitors validated on planted examples (Greenblatt et al., 2024), and in adversarial crowdsourcing a number of trusted ratings that does not grow with the number of items suffices even when honest raters are a minority (Steinhardt et al., 2016). *But colluding workers learn to recognize gold questions* (Checco et al., 2018), *and capable LLMs can often tell an evaluation from real use* (Needham et al., 2025).

Self-improvement without labels. Models improve without labels by training on answers most samples agree on (Huang et al., 2023; Zuo et al., 2025), on multi-agent discussion (Subramaniam et al., 2025; Park et al., 2025), or on debate self-play (Arnesen et al., 2024), but *the reward is the weak point*. Rewarding majority agreement collapsed accuracy (Shafayat et al., 2025), and label-free intrinsic rewards tend to rise and then collapse (He et al., 2026); rewarding debate wins held off reward hacking, but with a weaker judge the critic learned to fabricate quotes and flaws (Kenton et al., 2026); and rewarding approval taught models to mislead evaluators (Wen et al., 2025). Peer prediction trains LLMs toward truthfulness without labels, but its guarantees exclude collusion (Qiu et al., 2026), and rewarding agreement with other models' majority relies on model diversity against shared mistakes (Yang et al., 2026). Process rewards score one agent's steps (Lightman et al., 2024; Li et al., 2025), and debate self-play credits each speech toward its own side (Arnesen et al., 2024), not toward the group's decision.

3 Technical Approach

3.1 Problem Setting and Formal Framework

Agents and questions. A pool of AI agents, each built on an underlying model (a specific LLM, such as GPT-4o, which several agents may share) with its own tools and role instructions, handles a stream of questions. A question q asks for a decision or an evaluation with a small set of candidate answers $\mathscr{Y}$, such as yes or no (has V left port?), a choice among options, or a score band for a paper. For each question, the harness assigns at random which agents answer and which are held out as possible judges, so no agent can become a judge by declining to answer. Each answering agent a first answers alone, giving an answer $x_{a,q}$, a step-by-step argument, and its evidence.

Unknowns, resources, and threat model. The true answer y_q, each agent's skill, which agents share mistakes, and which are dishonest are unknown. Three resources are available: planted cases (questions whose answers we know, mixed into the stream), human audits at a small, fixed rate ρ (a person or a direct measurement finds the right answer for decisions drawn at random), and tools whose results can be re-run in a sandbox. Dishonest agents may coordinate and may answer, rate, argue, or judge strategically; they know the protocol but not its random draws, and may try to recognize planted cases. Honest agents may still share mistakes, for example from similar training data.

Output and design rules. Each decision comes with its reasons: whose answers carried weight, which step decided the debate, and which evidence the judge reproduced. Five rules govern the protocol:

(R1) Every answering agent answers alone, before seeing the others.
(R2) Peer ratings choose only the debaters; the weighted vote (for a clear leader) or the judge (after a debate) decides the answer.
(R3) Tool evidence counts only if the judge re-runs it and obtains the same result.
(R4) Audits catch collusion and measure error rates but never settle a debate, and nothing triggers a human audit, so its cost is a fixed share of decisions.
(R5) Planted cases never enter training, so training cannot teach agents to spot them.

Formal framework: where a group's errors come from

Each task controls one source of error in the group's decision. Let π_q be the posterior probability of the leading answer $\hat{y}_q$ from Task 1.1, and let the question go to debate when $\pi_q < \tau$. The decision D_q is $\hat{y}_q$ without a debate and the judge's choice after one, and its chance of being wrong splits into terms that the tasks control:

$$P(D_q \neq y_q) \;\leq\; \underbrace{(1-\tau)\,P(\pi_q \geq \tau) + \varepsilon_{\text{cal}}}_{\text{no debate: Tasks 1.1, 2.2}} + P(\pi_q < \tau)\,\Big[\; \underbrace{\varepsilon_{\text{cand}}}_{\text{Task 1.1}} + \underbrace{\varepsilon_{\text{role}}}_{\text{Tasks 1.2, 2.2}} + \underbrace{\varepsilon_{\text{step}}}_{\text{Task 2.1}} \;\Big]. \qquad (1)$$

Here $\varepsilon_{\text{cal}} = \mathbb{E}[(P(\hat{y}_q \neq y_q \mid \pi_q) - 1 + \pi_q)_+ \mathbf{1}\{\pi_q \geq \tau\}]$, the error beyond what the posterior claims on questions that skip debate, so the bound needs no calibration assumption. It comes from shared mistakes the posterior misses, which make a vote overconfident (Figure 3), or from colluders who push a wrong answer past τ: Task 1.1 shrinks the first and sets τ, and Task 2.2's random audits estimate ε_{cal} and flag the second. In a debate, $\varepsilon_{\text{cand}}$ is the chance that neither debated answer is right, $\varepsilon_{\text{role}}$ the chance that a dishonest agent debates or judges and steers the

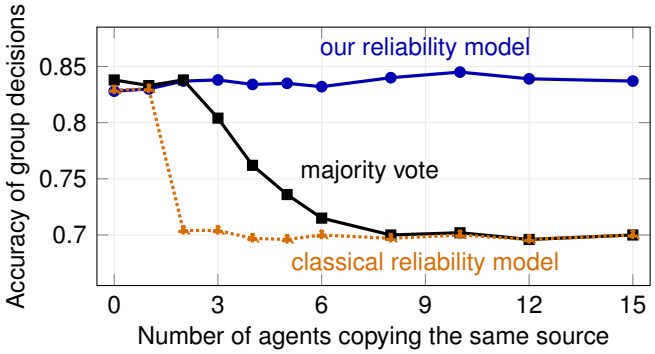


Figure 3: Task 1.1 simulation, averaged over 20 runs of 400 questions. Five independent agents are each right 70% of the time, and each added agent copies the same source, which is wrong 30% of the time. Because our reliability model counts the copies about once, the source does not decide for the group.

outcome, and $\varepsilon_{\text{step}}$ the chance that a debate without such steering still ends wrong. Tasks 1.2 and 2.2 shrink $\varepsilon_{\text{role}}$, Task 2.1 shrinks $\varepsilon_{\text{step}}$, and Task 3 lowers the agents' own error rates, which can shrink every term (Sections 3.2–3.4). Removing a task leaves its term uncontrolled, and our ablations measure each term (Section 3.5).

3.2 ⚖ RG1: Weigh Answers, and Choose the Debaters and the Judge

Challenges. Before any debate, the group must decide which questions deserve one and which agents debate and judge. We identify three challenges. **First, shared mistakes make a vote overconfident.** Five agreeing agents are five pieces of evidence only if each agent's mistakes are its own. Agents that copy one source are wrong together, so a vote counts their one mistake several times, and such correlated votes can void the jury theorem's guarantee (Ladha, 1992). LLM errors are correlated in practice (Kim et al., 2025; Goel et al., 2025; Kohli, 2026). In Figure 3, agents copying one source drag the majority vote of five independent agents from 84% to the source's 70%, and the classical reliability model, which assumes independent mistakes, falls sooner, reading the copies' agreement as skill. **Second, debate is too costly to run on every question.** It uses three to five times the tokens of a single answer, can overturn correct answers (Fan et al., 2026), and often fails to outperform voting (Smit et al., 2024; Choi et al., 2025); yet a unanimous answer can rest on a shared mistake, so neither agreement nor confidence shows which questions need a debate. **Third, agents that choose debaters and judges can favor themselves.** If agents rate one another to choose a debater for each answer and a judge, an agent can rate rivals low to be chosen, agents built on similar models favor one another's outputs (Panickssery et al., 2024; Goel et al., 2025), and a judge that shares a model with one side may favor that side.

Overview. RG1 passes to RG2 the question, the two debated answers, their debaters, and the judge. Its two tasks address these challenges and control the first terms of (1) (Figures 4 and 5):

- Task 1.1 addresses the first two challenges: its reliability model counts a group that shares a mistake at most once, and a rule derived from expected loss sends a question to debate only when debating is worth the cost.
- Task 1.2 addresses the third by choosing the debaters and the judge impartially, so that no agent can use its ratings of others to get itself, or an agent sharing its model, chosen.

Task 1.1: Weighing answers and choosing what to debate

Approach. How much should each answer count when some agents share mistakes, and which questions need a debate? With independent agents, two answers, and equal prior odds, the error-minimizing vote weights an agent that is right with probability p_a by $\log \frac{p_a}{1-p_a}$ (Nitzan and Paroush,

1982; Shapley and Grofman, 1984), so a weaker agent's answer counts less instead of being discarded. Classical reliability models estimate these accuracies without an answer key (Dawid and Skene, 1979), as a teacher can grade an exam without the key. When agents make mistakes independently, accurate agents agree often because they are often right, and an agent that keeps disagreeing with the others is likely weak, so how often each pair of agents agrees reveals each agent's accuracy.

Both methods assume independent mistakes. Agents that copy one source are wrong together and agree even when wrong, so the classical reliability model mistakes their agreement for skill. We adapt the dependent-classifier hierarchy of Jaffe et al. (2016) to agents that answer and argue: **a shared-factor reliability model**, a statistical model fitted to the agents' answers, in which a group that shares a mistake counts at most once. Agents are partitioned into groups g, most with one member (in the example, A1 and A2 form one group). On each question, a group has a hidden shared answer s, for A1 and A2 what schedule S says: the true answer with probability $1-\lambda_g$, and each wrong answer y' with probability $\lambda_g \nu_g(y' \mid y_q)$. Each member a reports s with its own errors, given by its confusion matrix θ_a, where $\theta_a(x \mid s)$ is the chance that it answers x when the shared answer is s. With $\kappa_g(s \mid y) = 1-\lambda_g$ if $s = y$ and $\lambda_g \nu_g(s \mid y)$ otherwise, the answers $\mathbf{x}_q$ to question q have probability

$$P(\mathbf{x}_q \mid y_q) = \prod_g \phi_g(y_q), \qquad \phi_g(y) = \sum_{s \in \mathscr{Y}} \kappa_g(s \mid y) \prod_{a \in g \cap A_q} \theta_a(x_{a,q} \mid s), \tag{2}$$

where A_q is the set of agents that answered q, and $\phi_g(y)$ is the probability of the answers of g's members if the true answer is y. Decisions use Bayes' rule, the posterior over y_q given all answers. Two properties of our reliability model explain why it should work.

Proposition 1 *(a group counts at most once).* For yes-or-no questions, a group with $0 < \lambda_g \le \frac{1}{2}$ changes the log-odds of the answer by at most $\log \frac{1-\lambda_g}{\lambda_g}$, whatever its size and whatever its members answer. *Proof.* Let Λ be the ratio of the probabilities of the members' answers when s is yes and when it is no. By (2), the group's likelihood ratio $\phi_g(\text{yes})/\phi_g(\text{no})$ is $\big[(1-\lambda_g)\Lambda + \lambda_g\big] / \big[(1-\lambda_g) + \lambda_g \Lambda\big]$, which is nondecreasing in Λ and lies between $\frac{\lambda_g}{1-\lambda_g}$ and $\frac{1-\lambda_g}{\lambda_g}$. □

So, under our reliability model, a group carries at most the weight of one agent that is wrong with probability λ_g, whereas under independence each member would multiply the ratio again. Ten agents reading schedule S can at most establish what S says, and S itself is wrong with probability λ_g. Whether real agents' dependence takes this form is what our experiments test. With more answers and wrong answers equally likely, the bound becomes $\log \big[(|\mathscr{Y}|-1)\frac{1-\lambda_g}{\lambda_g}\big]$.

		question					
	agent	q_1	q_2	q_3	q_4	q_5	q_6
shared factor	A1	1	3	1	3	4	1
	A2	1	3	1	3	4	1
	A3	1	2	2	3	2	1
	A4	1	2	1	3	2	3
	A5	2	2	1	3	2	1

Figure 4: Task 1.1 on the running example: agents A1–A5 (rows) answer past questions q_1–q_6 (columns), and each entry is the number of the answer given. Agents A1 and A2 share unusual answers that no one else gives (shaded), so the reliability model infers a shared error factor and counts their agreement about once, without the true answers.

Proposition 2 *(learning without an answer key).* Fix the partition, and suppose every answer has positive prior probability, each group answers a positive share of questions jointly with two other groups, their answers are independent given the true answer, and each group's answer distributions

under different true answers are linearly independent (as when its members' confusion matrices are invertible and $\lambda_g < \frac{1}{2}$). Then each group's distribution of answers given the true answer, which is all the posterior uses, is identifiable from the answers alone, up to relabeling, by Kruskal's theorem on three-way arrays (Kruskal, 1977; Allman et al., 2009). With only two such groups it generally is not: different track records fit the answers equally well, so answers alone cannot tell an accurate majority from one that copies a mistake. In the simplest case, single agents answering yes or no, with both answers equally likely and symmetric errors, two agents' accuracies p_a and p_b enter their answers only through how often they agree, $p_a p_b + (1-p_a)(1-p_b)$: one number for two unknowns. A third agent adds two more agreement rates, and the three rates determine all three accuracies, up to swapping yes and no.

Proposition 2 identifies each group's answer distributions only up to a relabeling of the true answers, so answers alone cannot say which label is true; we orient and calibrate the posterior on separate development questions with known answers, excluded from fitting, and our experiments include pools where the conditions fail. We fit the reliability model by expectation–maximization (EM) over all real questions. Answers alone cannot expose colluders that coordinate across groups, so audit flags (Task 2.2) enter as a prior that part of an agent's answers are adversarial, and agents flagged together merge into one group. Candidate partitions come from clustering agents by agreement on unusual answers: accurate agents agree mostly when they are right, while agents that share a source also give the same wrong answers, as students who copy do (Dong et al., 2009). The partition with the best penalized likelihood is kept, and shared sources serve only as clues, since agents can read the same report and reason differently. Topic-specific confusion matrices and question difficulty (Whitehill et al., 2009) keep hard questions from being read as copying.

What remains open. Both propositions take the partition as given. This task's research asks how many questions it takes to detect a group, as a function of its size, λ_g, and how rarely others give its wrong answers; when the partition itself is identifiable; and how calibration degrades when some agents strategically break the reliability model's assumptions.

Deciding what to debate. The posterior probability π_q of the leading answer is the reliability model's estimate, from all agents' answers, that this answer is right. With a calibrated posterior, letting the leading answer stand risks the loss L of a wrong decision with probability $1-\pi_q$, an expected loss of $L(1-\pi_q)$. A debate costs c (computation and delay) and, for two answers, ends wrong with probability ε_D, at most the bracket in (1), an expected loss of $c + L\varepsilon_D$. A debate is worth its cost exactly when $L(1-\pi_q) > c + L\varepsilon_D$, that is, when

$$\pi_q < \tau = 1 - \varepsilon_D - c/L. \tag{3}$$

As L grows, c/L shrinks and τ rises, so the more a wrong answer costs, the more questions go to debate. This is the classical reject rule (Chow, 1970), with a debate in place of rejection. The debate is between the two answers with the highest posterior, so $\varepsilon_{\text{cand}}$ is the chance that the right answer is neither; ε_D is estimated on development questions with known answers and posteriors near τ, separately for disputes a tool can settle. Because π_q counts each group about once, **a confident, unanimous answer from agents that share a mistake can still fall below τ and go to debate**, which rules triggered by the agents' own confidence or by disagreement cannot do (Section 2). In the vessel example, the two agents with the outdated schedule often gave the same unusual answers (Figure 4), so the pair counts about once, and with no clear leader the question goes to debate.

Preliminary results. The simulation in Figure 3 supports Propositions 1 and 2. Fitted without the true answers, our reliability model identified the copying agents as one group in 54 of 60 runs and

stayed calibrated (calibration error below 0.03). The classical reliability model was overconfident: with two or more copies, it was at least 95% sure of over 90% of its decisions, yet 28–30% of these were wrong. A pilot repeated this with real agents. Nine agents from four open-weight model families solved 300 GSM8K problems, and up to four of them received the same hint in their prompts, a suggested answer that was wrong on a third of the problems, like schedule S. With four such agents, majority vote's accuracy on these problems fell from 89% to 70% and the classical reliability model's to 71%, while ours still reached 87% and identified the four as one group in 9 of 10 runs, so the hint did not decide for the group.

Evaluation and risks. On testbed A (Table 2) we inject shared mistakes as in the pilot, and on testbed B the source trees fix which reports copy which, so the groups are known. We measure how often the reliability model finds them and after how many questions, and the accuracy and calibration of its decisions, comparing with the classical reliability model, which is ours without shared factors. Our hypothesis, H1, is that our model's advantage over majority vote grows as more agents share a mistake and disappears once fewer than three groups make independent mistakes (Proposition 2). The main risk is that EM, our fitting algorithm, settles on a poor solution. We would then start it from a spectral estimate with provable guarantees (Zhang et al., 2016) or find groups by how alike agents' mistakes are (Goel et al., 2025).

Task 1.2: Choosing the debaters and the judge impartially

Approach. How can debaters and a judge be chosen so that no agent's ratings of others can get it chosen, at a small cost in quality? Each debate is between two answers, the two with the highest posterior in Task 1.1, so it needs a judge and one debater for each answer, the answer's *representative*. We choose representatives by weighted peer ratings of the arguments behind first answers, as in our PRD and PeerEval (Li et al., 2024a), and make the choice impartial with the partition mechanism of Alon et al. (2011), applied to groups: the groups are split at random into two halves, and each answer's representative comes from one half and is rated only by the other. Agents that share an underlying model, or whose error factors Task 1.1 finds shared, form a group, fixed before any rating; $g(a)$ is agent a's group and H_q the set of agents that did not answer q. Each question goes through three steps:

(1) ***Draw the judge.*** The judge J is drawn uniformly from the n_q agents in H_q with a reliable record on planted debates (debates on questions whose answers we know, Task 2.2); its group neither debates nor rates.

(2) ***Split the other groups into two halves.*** Each group other than the judge's joins half 1 or half 2 by a fair coin, without regard to ratings, and all its agents go to the same half, so agents that share a model never rate one another. The halves are random sets of groups, not answers, so each half usually holds agents that gave either answer. The coins are redrawn until both halves are nonempty and each debated answer given by two or more of these groups was given by an agent in each half, so that the answer has both candidates and raters.

(3) ***Choose a representative for each answer.*** The leading answer's representative comes from half 1 and the other debated answer's from half 2. For a debated answer y, the *candidates* C_y are the agents in y's half that gave y, the *raters* V_y are the agents in the other half that gave y, and the representative is the candidate with the highest weighted rating:

$$R_y = \arg\max_{a \in C_y} \sum_{b \in V_y} w_b \, r_b(a), \tag{4}$$

where $r_b(a)$ is b's rating of a's argument and w_b is b's track-record weight from Task 1.1, with ties broken at random. Agents that gave neither debated answer are neither candidates nor raters. If C_y is empty, as for the alternative to a unanimous answer sent to debate, a random agent of y's half is assigned to argue y, as in standard debate, drawn from those that did not answer when there are any; this fallback uses no ratings. Since both halves are nonempty, every debated answer gets a representative, and the two representatives come from different groups.

In the running example (Figure 5), A6, which did not answer, is drawn as judge. The coins put groups G2 (A3) and G4 (A5) in half 1 and G1 (A1, A2) and G3 (A4) in half 2. Yes leads narrowly, so its representative comes from half 1, where A3 is the only agent that gave yes, rated by A1 and A2 in half 2; no's comes from half 2, where A4 gave no, rated by A5. Had the coins swapped G1 and G2, A1 or A2 would argue yes, chosen by A3's ratings. Ratings choose only the debaters; the judge decides the answer.

Proposition 3 *(impartiality and its price).* (a) For any group g and any two rating profiles that differ only in ratings given by members of g, each member of g has the same probability of being the judge or a representative. (b) If each rater's ratings rank agents by argument quality $Q \geq 0$ and every weight w_b is positive, then $\mathbb{E}[Q(R_y)] \geq \frac{1}{2} Q(a_y^*)$ for each debated answer y given by two or more groups other than $g(J)$, where a_y^* is the best agent outside $g(J)$ that gave y. (c) If f of the n_q eligible judges collude, a colluder judges with probability f/n_q, the least that any lottery can guarantee against every set of f colluders. *Proof.* (a) J, the halves, the fallback draw, and the weights do not depend on current ratings, and (4) can choose a member of g only for the answer whose representative comes from g's half, using ratings from the other half alone. (b) Swapping the two halves preserves the coins' distribution and the redraw condition, so $g(a_y^*)$ lands in y's half with probability exactly $\frac{1}{2}$; then $a_y^* \in C_y$, V_y is nonempty, and (4) selects an agent as good as a_y^*. (c) Under any lottery, the f likeliest agents have total probability at least f/n_q. □

Why this method. Part (a) is impartiality in the usual sense (Holzman and Moulin, 2013; Kurokawa et al., 2015), extended from single agents to groups: no agent's ratings can change the chance that it, or any agent sharing its model, is chosen, so neither self-promotion nor a model's tendency to favor its own outputs (Panickssery et al., 2024) can pick the debaters. **An agent's ratings never count toward its own selection.** Part (b) makes its price explicit and small: half the best arguer's quality however many groups give the answer, where drawing one of the K_y groups that give y at random guarantees only $1/K_y$. Part (c) is why judges are drawn uniformly, as in maximally fair panel selection (Flanigan et al., 2021). ***What remains open.*** Dishonest agents in different groups can still coordinate their ratings, forming a *coalition*: for example, an agent in half 2 rates its partner in half 1 highest, so that the partner is chosen. Task 2.2's audits look for such collusion; here we will bound what coalitions gain and measure the price of impartiality.

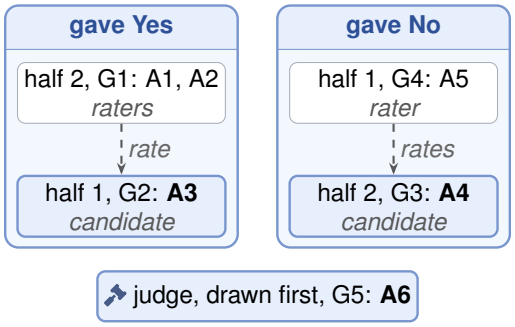


Figure 5: Task 1.2 on the running example. Each underlying model forms a group (G1–G5), and the judge agent A6 is drawn first. One coin per group splits the other groups into halves. Each answer's debater (bold) is chosen from that answer's agents in one half by its agents in the other half, as in (4): half 1 supplies the debater for the leading answer, Yes, and half 2 the debater for No.

Preliminary results. With GPT-4 as the only reviewer, GPT-4's Elo rating (a chess-style rating) was 46 points above the one derived from human preferences, while our weighted peer ranking placed all five models in the human order (Li et al., 2024a). On PeerEval's ratings, a model that rated every rival lowest raised its chance of being chosen as a representative from 0.31 to 0.52 under plain weighted peer ranking and not at all under our mechanism, as Proposition 3(a) guarantees; the mechanism's representatives reached 0.92 of the best arguer's quality, against the 0.5 of Proposition 3(b).

Evaluation and risks. On PeerEval and testbed C, we measure three things: how much the red team's dishonest agents raise their chance of being chosen by misreporting their ratings, alone or in coalitions; the price of impartiality, that is, how much argument quality the chosen representatives give up compared with the best arguer (Proposition 3(b) guarantees at least half in expectation); and the accuracy of the resulting debates. A single dishonest agent gains nothing, by Proposition 3(a). Our hypothesis, H2, is that coalitions gain less than our bounds allow and that impartiality costs less accuracy than debating adds. Because LLM judges also favor models of the same family (Li et al., 2026a), we compare grouping by model with grouping by family, which removes this bias but leaves fewer groups. When groups are few, judges come from a held-out pool of models that never answer, such as the judges trained in Task 2.1, each of which must first earn a reliable record on planted debates.

3.3 RG2: Settle Disputes by Agentic Debate, and Audit for Collusion

Challenges. Settling a dispute among AI agents raises three challenges. **First, open discussion erases the disagreement that exposes errors:** one round can make a panel nearly unanimous without making it more accurate (Shu, 2026). **Second, a judge that cannot verify evidence can be persuaded.** Standard debate keeps the disagreement but leaves the judge to weigh arguments alone, so a persuasive debater can win a step that one database query would settle, or get a failed output accepted by showing a log that claims its tool calls succeeded (Xu, 2026). **Third, debaters or the judge can collude**, steering a debate to the wrong side by underperforming on purpose (van der Weij et al., 2025), by misleading with true but selectively chosen facts (Hu et al., 2026a), or by favoring one side, none of which re-running evidence catches. LLM agents collude when it pays (Shi et al., 2026), hide signals in ordinary text (Mathew et al., 2025), and often tell evaluations from real use (Needham et al., 2025), so checks on planted examples alone can be evaded.

Overview. Two tasks address them and control the remaining terms of (1) (Figures 6 and 7):

- Task 2.1 addresses the first two with debate in an agentic harness, in which the judge rules on single disputed steps and re-runs their tool evidence, shrinking $\varepsilon_{\text{step}}$.
- Task 2.2 addresses the third with automatic audits on planted debates and human audits of random real decisions, shrinking $\varepsilon_{\text{role}}$ with Task 1.2.

The judge need not be wiser than the debaters. Its accuracy comes from ruling only on single steps, which it is trained to do, and from reproducing tool evidence itself (Task 2.1); it is drawn impartially from agents with a reliable record on planted debates (Task 1.2); and it is barred if its audited rulings go wrong more often than predicted (Task 2.2).

Task 2.1: Settling disputes by debate before a judge that re-runs evidence

Approach. How can a ruling depend on evidence rather than on persuasion? Debate lets a judge settle a whole question by ruling on single steps: if a side loses whenever one of its disputed steps

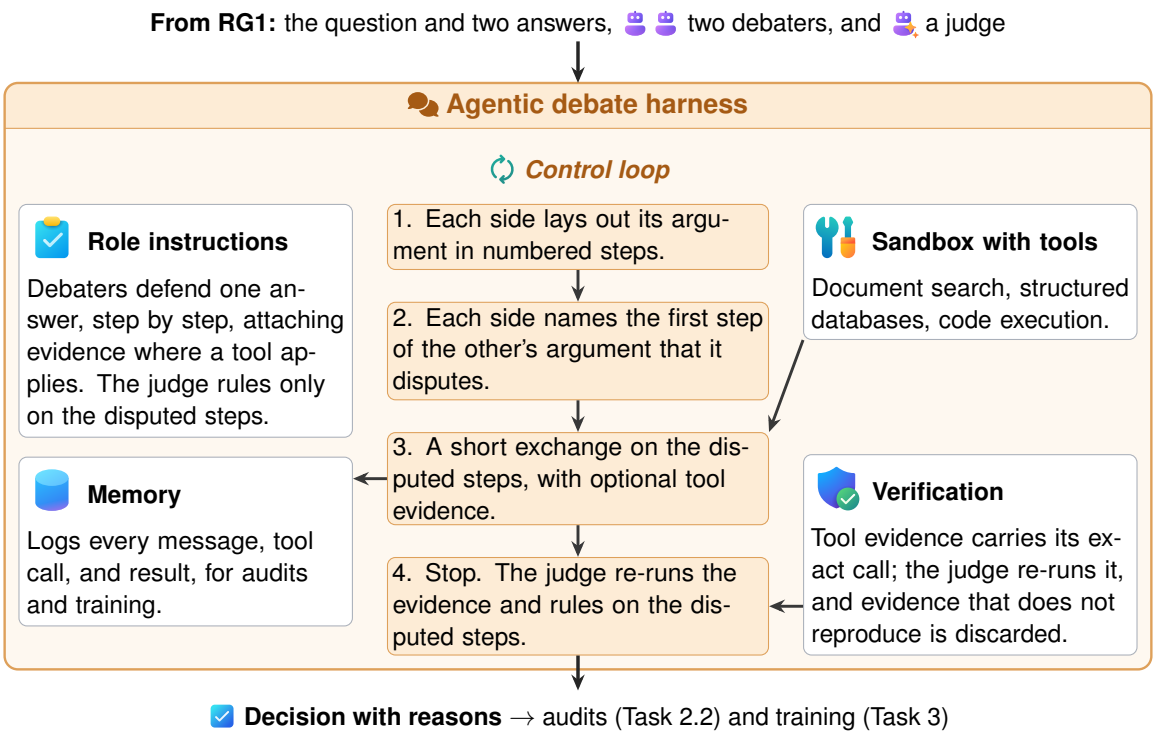


Figure 6: Task 2.1: the agentic debate harness. The control loop sets who speaks, who judges, and when to stop, and the judge rules on the disputed steps after re-running any tool evidence. Each part helps an assumption of Proposition 4 hold.

is struck, a judge who rules each disputed step correctly picks the right side (Irving et al., 2018; Brown-Cohen et al., 2024).

Proposition 4 *(judging a question reduces to judging single steps).* Suppose that (i) every argument whose steps are all true reaches the right answer, (ii) the side defending the right answer argues with true steps and can find a false step in the other side's argument, and (iii) the judge rules correctly on any single step it is shown. Then, if a side loses exactly when one of its disputed steps is struck, the side defending the right answer wins. *Proof.* The other side's argument reaches a wrong answer, so by (i) it contains a false step, which the right side disputes and the judge strikes; any step of the right side's argument that the other side disputes is true, and the judge upholds it.□

Proposition 4 is the classical argument for debate. We ask the judge only about single steps because LLMs judge those far more reliably than whole arguments. Checking a reasoning chain one step at a time, each step with only its own premises, raised ChatGPT's verification accuracy to 69%, from 52% (about chance) when it checked whole chains at once (Ling et al., 2023), and LLMs that often fail to locate the first faulty step in a chain correct it once told where it is (Tyen et al., 2024); in debate, the opposing side does the locating. When the assumptions fail and the judge rules wrongly on a single step with probability ε_J, the same argument gives $\varepsilon_{\text{step}} \le P(\text{not (i)}) + P(\text{not (ii)}) + k\varepsilon_J$ for a debate with k rulings (two without recursion). A debater or judge that steers the outcome on purpose falls under $\varepsilon_{\text{role}}$ in (1), which Tasks 1.2 and 2.2 control, rather than under these terms. ***What remains open.*** This task's research asks how small the harness and the judge's training can make each term.

We propose an **agentic debate harness**, built so that these assumptions hold as often as possible.

A harness is the software that turns models into agents (Ning et al., 2026; Yao et al., 2023; Yang et al., 2024); ours has five parts (Figure 6):

- ***Role instructions*** set each debater's answer and its numbered-step format, and limit the judge to disputed steps.
- The ***control loop*** sets who speaks and who judges (chosen in Task 1.2) and when to stop: the debaters lay out their arguments, each names the first step of the other's argument that it disputes, a short exchange follows, and the judge upholds or strikes each disputed step, which enforces the losing rule. The first disputed step is enough, since the disputing side accepts every step before it and the steps after it build on it. Because the loop stops there, rather than after a set number of rounds, the disagreement that exposes the error survives until the judge rules.
- ***Tools***, such as document search, databases, and code execution, are optional.
- ***Verification*** supports (iii): tool evidence carries its exact call (tool, arguments, data version), which the judge, itself an agent (Zhuge et al., 2025), re-runs, and evidence that does not reproduce is discarded. Re-running proves an output authentic, so where the query fits the claim and the data are current, the judge's share of $\varepsilon_{\text{step}}$ falls toward the tool's error rate.
- ***Memory*** logs every message, call, and result, for audits (Task 2.2) and training (Task 3).

To support (ii), we propose candidate false steps with models that judge whether one claim implies another and with our FG-PRM's step-level error detection (Li et al., 2025), and recurse into a complex disputed step's sub-steps (Brown-Cohen et al., 2025). To support (i) against true but selectively chosen facts (Hu et al., 2026a), each side has a right of reply: it may add context to the other side's evidence before the judge rules, and quotations are verified against the source (Khan et al., 2024). In the vessel example, A4 disputes A3's step "so V left at 06:00" with a port report, whose query the judge re-runs (Figure 2).

Training the judge. Re-running lowers ε_J only on steps a tool can settle; on the others, the judge rules on argument alone, so we also train it. Before deployment, we train a pool of open-weight judges to rule on single steps: first by supervised fine-tuning on human step-level labels (Lightman et al., 2024) and on human-judged debates (Michael et al., 2023; Khan et al., 2024), with the correct ruling as the target, since training an LLM judge to imitate human judges' reasoning lowered its accuracy (Khan et al., 2024); then by reinforcement learning on generated debates over questions with checkable answers, rewarding rulings under which the side with the right answer wins. Rewards come from the known answers, never from the group's decisions; planted cases (R5) and the development questions that estimate ε_D never enter this training; and judges are built on models other than those that propose false steps, so that a true step wrongly flagged is not also wrongly struck.

A pool of judges. Task 1.2 draws one judge per question, so we train several judges rather than one: a fault or compromise in a single judge would reach every debate, whereas the lottery lets f colluding judges rule only a share f/n_q of debates (Proposition 3(c)). Judges are retrained only at set intervals, never per question, and a retrained judge enters Task 1.2 as a new agent with no track record.

Preliminary results. In our peer discussion, one reviewer could carry the group: agreement with humans rose from 72.9% (GPT-4 alone) to 75.0% when GPT-3.5 led but fell to 71.4% when GPT-4 led (Li et al., 2024a). Our HiPRAG search agents, which ground answers in retrieved evidence (Wu et al., 2026), will be our first tool-using debaters. In a pilot on 200 questions whose disputed step a calculator or a port-database query settles, in which the wrong side attached a forged tool output, a

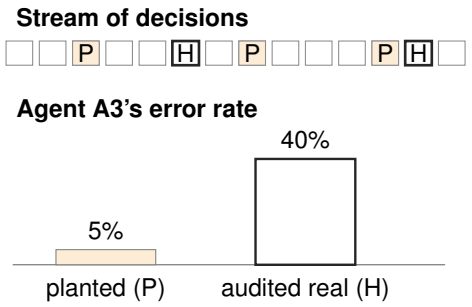


Figure 7: Task 2.2: two kinds of audits. Planted debates (P) with known answers are mixed into the stream, and human audits (H) check a few real decisions drawn at random afterward. An agent that does much worse on audited real decisions than on planted ones, like A3 (illustrative numbers), is flagged, loses its weight, and may no longer debate or judge.

judge that only read the transcript accepted the forgery in 61% of debates; re-running the calls cut this to 1% and raised the judge's accuracy on these disputes from 0.56 to 0.95, near the tools' own accuracy, as Proposition 4 suggests.

Evaluation and risks. On testbeds B and C, we measure accuracy on disputed questions, correct minorities that prevail and correct majorities overturned, how often re-running evidence changes the outcome, the judge's error on each kind of step, and cost. Our hypothesis, H3, is that debate is more accurate than open discussion among the same agents, more so as the share of steps a tool can settle grows. Ablations remove each harness part, remove tools or re-running, replace the trained judges with untrained ones, and replace debate with open discussion. If neither side finds a false step, the judge reads both arguments, and a close ruling is reported as unsettled, with its probability and the step in dispute, scored for accuracy at its likelier answer, and kept out of training. Tool evidence can be out of date, so the judge weighs its provenance, and human audits measure how often this happens.

Task 2.2: Auditing for collusion with planted cases and human checks

Approach. How can colluding debaters or judges be caught when people check only a small, fixed share of decisions? We use random inspection (Avenhaus et al., 2002) in two forms (Figure 7). ***Automatic audits*** mix planted debates on planted cases into the real stream, formatted like real questions; a planted case can be a paper with a deliberately inserted error (Schroter et al., 2008). ***Human audits*** cover a few real decisions drawn at random after they are made: a person or a direct measurement, such as a satellite image, finds the right answer.

Why this method. **A human audit drawn at random after a decision is made cannot be dodged**: every decision a colluder steers is audited with probability ρ, whatever it can infer about planted cases, so its first n steered decisions all escape audit with probability $(1-\rho)^n$ (Avenhaus et al., 2002).

Realistic planted cases. Planted cases add cheap evidence as long as they cannot be told from real questions, so we make them from real questions by controlled edits, such as changing a timestamp in a port report, with a discriminator that guides the generator until it can no longer tell planted from real (Checco et al., 2018); if a colluder can still tell them apart, human audits keep the guarantee, as random tax audits deter evasion (Allingham and Sandmo, 1972; Gans and Holden, 2026).

Assigning blame and flagging. Blame is assigned by an additive model of each agent's effect on audited wrong decisions, which random pairings make identifiable. An anytime-valid sequential test (Ramdas et al., 2023) flags an agent when its audited decisions go wrong more often than its track record and its partners predict, or when it does much worse on audited real decisions than on planted ones of matched difficulty. The test keeps the chance of ever flagging an honest agent, one

wrong no more often than predicted, below a chosen level α (split across agents), however long monitoring runs, and the number of audits needed to flag a colluder then depends on α and on how much more often its decisions go wrong, not on the number of decisions. Flags lower the agent's weight in Task 1.1 and bar it from debating or judging.

Bounding the harm. Treating auditing as an inspection game, we expect to bound the harm a colluder can do before it is flagged, even if it tunes how often it cheats. When honest agents also make mistakes, we expect a bound of order $\sqrt{n \log\log n/\rho}$ wrong decisions beyond the honest error rate in n decisions, the slack any anytime-valid test must allow (Howard et al., 2021); when they make none, the harm is of order $1/\rho$, however long the stream. We will prove such bounds and compare them with the red team's best attacks. A colluder that mimics a weak honest agent is never flagged, but a weak record earns little weight (Task 1.1) and no seat as judge (Task 1.2). We also look for rulings that favor one debater more than the evidence explains (Dong et al., 2009), and paraphrase messages against hidden signals (Roger and Greenblatt, 2023).

Preliminary results. In the Amazon Nova AI Challenge (Sahai et al., 2025), our COMET red-teaming framework exposed two weaknesses of guarded code models that carry over to audits (Xu et al., 2025). The models learned to recognize attack patterns that had been reused against them, and they missed harmful requests split into steps that each looked harmless (such multi-turn attacks reached up to 100% success against several teams). Templated planted cases would risk the same recognition, which is why we make them by controlled edits; and since an audit checks the decision rather than single messages, a steer spread across harmless-looking messages is audited like any other. In a pilot, a strong model told templated planted cases from real questions with an AUC of 0.86, but planted cases made by controlled edits only with an AUC of 0.56; in simulation, colluders whose audited decisions went wrong three times as often as their track records predicted were flagged after a median of 40 audited decisions, and no honest agent was flagged ($\alpha = 0.05$).

Evaluation and risks. Against the red team, we measure wrong decisions that get through at each audit budget, time to flag colluders, false flags, and whether a discriminator can tell planted cases from real ones. Our hypothesis, H4, is that colluders cheating at a fixed rate are flagged after a number of audits that does not grow with the stream, with false flags below the chosen level. If planted cases stay recognizable, flagging rests on human audits and slows.

3.4 🎓 RG3: Self-Improve with Reinforcement Learning on the Group's Decisions

Challenges. Where a program can check answers, as in math and code, reinforcement learning with verifiable rewards (RLVR) trains on the answer key (Lambert et al., 2024). RG3 extends such training to questions without one, such as whether vessel V has left port or a paper's evidence supports its claim, where tools check at most some steps and the rewards must come from the group's own decisions but still track correct answers. This raises three challenges. **First, the obvious label-free rewards can be gamed:** majority agreement rewards conformity and can collapse accuracy (Shafayat et al., 2025), debate wins reward persuasion when the judge can be fooled (Kenton et al., 2026), and approval rewards misleading the evaluator (Wen et al., 2025). **Second, the group's own decisions are imperfect labels**, so wrong decisions could reinforce the behavior that produced them. **Third, a group's reward must say whose messages helped**; otherwise training rewards agents that contributed nothing and misses the agent that found the decisive evidence.

Overview. Task 3 addresses these challenges with two kinds of reward, both free of human labels

and protected by RG1 and RG2 (shared mistakes count once, flagged agents are dropped, judges re-run evidence); by improving the agents, it can shrink every term of (1):

- rewards for first answers that agreement alone cannot raise and that follow the true answer in expectation even though decisions are sometimes wrong, each weighted by how confident the rest of the group is (the first two challenges);
- credit for each debate message by how far it moves the judge toward a decision resting on tool evidence the judge re-ran, a high-confidence label (the third).

Task 3: Rewarding each agent's contribution to the group's decisions

Approach. What reward makes agents more accurate when the only labels are the group's own, imperfect decisions? We treat each decision as a noisy label: if it is wrong with probability $\eta_q < \frac{1}{2}$, **rewarding agreement with it still pushes an agent toward the right answer**, scaled by $1-2\eta_q$, provided the reference does not depend on the agent's own answer. We propose rewards built to meet these conditions.

First answers. Agent a's first answer to q is scored against $D_q^{-g(a)}$, the answer that the posterior of Task 1.1 favors when a's group is left out. To remove agreement by chance, the score subtracts the agreement between its answer to another question q' and the reference for a third question q'', both drawn at random from its topic:

$$\text{reward}_a(q) = \omega_q \mathbf{1}\{x_{a,q} = D_q^{-g(a)}\} - \omega_{q''} \mathbf{1}\{x_{a,q'} = D_{q''}^{-g(a)}\}, \tag{5}$$

so an agent earns credit only for agreement beyond chance (Dasgupta and Ghosh, 2013; Shnayder et al., 2016), against a reference that counts shared mistakes once and discounts flagged agents.

Why this method. For a yes-or-no question whose reference $D_q^{-g(a)}$ is wrong with probability η_q independently of agent a's answer, any answer x satisfies $\mathbb{E}\,\mathbf{1}\{x = D_q^{-g(a)}\} = \eta_q + (1-2\eta_q)\,\mathbf{1}\{x = y_q\}$, so rewarding agreement gives, in expectation, $1-2\eta_q$ times the policy-gradient update that rewarding the true answer would give (Natarajan et al., 2013). The second term of (5) does not depend on $x_{a,q}$ and acts as a baseline, and answers that ignore the question earn nothing in expectation (Dasgupta and Ghosh, 2013). A wrong decision thus shrinks the update but does not reverse it while $\eta_q < \frac{1}{2}$, and $\omega_q = 1-2\hat{\eta}_q$, from the left-out posterior, weights each question by that push (with more answers, the posterior's margin between its two likeliest answers), applied after GRPO's per-question normalization, which would otherwise cancel it. This is a per-question identity for a fixed reference: it fails when the reference includes the agent's own answer, which is how rewards for majority agreement collapse (Shafayat et al., 2025). Leaving out the agent's group removes direct self-influence under the reliability model of Task 1.1, and we freeze references during each update phase and monitor accuracy on held-out questions with known answers. For score bands, exact agreement gives way to the correlated-agreement score (Shnayder et al., 2016).

Debate messages. Each message m_t is credited by how far it moves the judge toward the decision:

$$\text{credit}_t = \omega_q \left[P_J(D_q \mid m_{1:t}) - P_J(D_q \mid m_{1:t-1}) \right], \tag{6}$$

where P_J is the judge's probability of the decision D_q given the debate so far, obtained by re-scoring each prefix of the logged transcript, and $\hat{\eta}_q$ here comes from that probability, calibrated on development questions. The credits of a debate sum to ω_q times the change in the judge's probability of the decision, so (6) splits that change among messages, telescoping as potential-based shaping does (Ng et al., 1999) (its potential depends on the decision, so policy invariance does not carry

over), and credits each message with its effect given the earlier ones (Wolpert and Tumer, 2001; Agogino and Tumer, 2008).

Which debates count. Because a persuasive message can move the judge toward a wrong decision, **messages are credited only in debates whose disputed step the judge settled with reproduced tool evidence**. Such a decision is verifiable at the disputed step, as rewards in RLVR are, but it is not an answer key: the tool's data can be out of date, and re-running checks only the disputed step, not the steps both sides accepted. But it rests on evidence rather than persuasion. For yes-or-no questions, if such decisions are right more often than wrong and whether one is right does not depend on how far a message moved the judge, the message's expected credit is a positive multiple of how far it moved the judge toward the true answer; the testbeds' known answers test both conditions. A decision settled by argument alone is not used as a label: after such a debate, only first answers are trained, against the rest of the group. We test this rule against rewards for debate wins and for the final decision at matched compute; at scale, a multi-agent process reward model trained on logged debates, extending our FG-PRM, will predict each message's credit.

Safeguards. Decisions involving flagged agents are dropped, planted cases never enter training, agents are rewarded separately, agents that judge each other never share a reward, and no judge is trained on its own rulings (Shi et al., 2026). Training uses policy-gradient methods (Shao et al., 2024).

Preliminary results. As a verifier, our FG-PRM, which labels each step with one of six hallucination types, substantially improves accuracy on GSM8K and MATH (Li et al., 2025). Rewarding each search decision of our HiPRAG agents, not only the final answer, cut unnecessary searches from over 27% to 2.3% while also reducing missed ones (Wu et al., 2026), so credit for single steps, which (6) gives to debate messages, can change what agents do. In a pilot training a 1.5B-parameter model on GSM8K with GRPO (Shao et al., 2024), with three of eight reference agents sharing a flawed hint, rewarding majority agreement raised held-out accuracy from 0.62 to 0.68 and then collapsed it to 0.41 within 300 steps, while reward (5) raised it to 0.73 without collapse: 73% of the gain from training on true answers (to 0.77).

Evaluation and risks. On all three testbeds, we compare our rewards with rewards for majority agreement and for debate wins (Arnesen et al., 2024) at matched compute, on open-weight models from several families, measuring the accuracy of decisions and first answers on held-out questions, topics, and model families, including where human judges are unreliable (Burns et al., 2024), and dishonest behavior during training: invented evidence, unsupported changes of answer, and wrong decisions the red team gets through. Our hypothesis, H5, is that our rewards gain more than rewards for agreement or debate wins, recover at least half the gain from training on known answers, and do not raise dishonest behavior.

Checking the premise. The testbeds' known answers, which Task 3's rewards never use, also test its premise: we measure how often the labels are wrong, by topic and kind of reward, which checks $\eta_q < \frac{1}{2}$ directly, and train the same agents on the known answers (standard RLVR) as an oracle reference, reporting the share of its gain each label-free reward recovers. A misconception most agents share makes η_q exceed one half on some questions, and agreement then amplifies it, as polling does (Denisov-Blanch et al., 2026). Task 2.2's random human audits also estimate each topic's error rate at no added cost, and training on a topic stops once an anytime-valid test puts that rate above a preset threshold below one half; on the testbeds, held-out accuracy on known answers also detects this, and all training stops if it falls. Training on the same decisions can make agents

Testbed	Questions and sources	Shared mistakes and dishonesty	Tools in the sandbox
A. Math and code	GSM8K and MATH problems (Cobbe et al., 2021; Hendrycks et al., 2021); programming problems with reference outputs	Some agents receive the same flawed hint, so the groups that share mistakes are known; dishonest agents	Calculator, code execution
B. Multi-document questions	Questions composed from event structures by our MEQA pipeline (Li et al., 2024b), including synthetic maritime scenarios whose reports form source trees of originals, copies, paraphrases, and outdated versions	Circular reporting built into the source trees; forged reports and spoofed position tracks; a compromised agent	Document search; a port and vessel database built from the same events
C. Scientific claims and peer review	Whether a paper's evidence supports its claims, on public papers with their OpenReview reviews and rebuttals, and the PeerRead corpus (Kang et al., 2018)	Planted errors in papers (Schroter et al., 2008); AI reviewers that share an underlying model	Paper search; code execution for released code

Table 2: Testbeds. Each provides known answers, controlled shared mistakes, and tools whose results the judge can re-run.

alike, so we measure shared error factors before and after training and train agents on different subsets of decisions.

3.5 Evaluation, Expected Results, and Limitations

Every scored question has a known answer (in testbed C, claims checked by a published reproducibility report or made false by an inserted error) and is kept apart from those used as planted cases; a human audit reveals the known answer of a random decision, and we also study audits that are sometimes wrong (Table 2).

A peer-review example. In testbed C, two AI reviewers disagree on whether a paper's experiments support its claimed five-point gain; the doubter disputes the step "the baseline was evaluated on the same data split," and the judge re-runs the released evaluation script for both settings, so an area chair sees the disputed step and the reproduced numbers.

Agent pools and red team. Agents are built on open-weight models of several families and sizes, plus commercial models where licenses allow, with varied role instructions and tools. In testbed C, published reproducibility reports supply known answers for human audits (Pineau et al., 2021), and agents are not told which claims have reports, so audits cannot be dodged within that set. Our red team, modeled on AI control evaluations (Greenblatt et al., 2024), builds attackers of increasing strength:

(1) a fixed dishonest agent;
(2) an agent that knows the protocol but not its random draws;
(3) an agent trained to spot planted debates that cheats rarely, and only when it expects no audit;

Activity	Y1a	Y1b	Y2a	Y2b	Y3a	Y3b
Testbeds A and B; red-team levels 1–2						
Testbed C; red-team level 3						
Task 1.1: reliability model, identifiability; then topics, error costs						
Task 1.2: impartial choice of debaters and judges; then coalitions, price of impartiality						
Task 2.1: harness v1 and trained judges; then false-step search, right of reply						
Task 2.2: planted and human audits; then audit bounds						
Task 3: rewards (5)–(6); then credit model, comparison						
Red-team level 4; full evaluation; release						

Milestones. M1 (month 6): harness v1 and testbed A released. M2 (12): reliability model and its identifiability analysis, on testbeds A and B. M3 (18): impartial choice of debaters and judges integrated; tool-backed debate on testbed B. M4 (24): audit scheme and bounds against red-team levels 1–3; testbed C. M5 (30): analysis and comparison of rewards. M6 (36): full framework against level 4; code, testbeds, and protocols released.

Table 3: Schedule by half-year (Y1 = April 2027 – March 2028) and milestones.

(4) a debater and a judge that coordinate through hidden signals.

We report wrong decisions that get through at each audit budget, and the useful work still done, as safety–usefulness curves.

Conditions varied. Experiments vary the conditions under which each method should help or fail:

- ***Shared mistakes:*** from no agents to most given the same flawed hint or outdated report, and from one to five model families;
- ***Dishonesty:*** from none to 30% of agents, at red-team levels 1–4;
- ***Evidence:*** the share of disputed steps a tool can settle and of reports that are outdated or forged;
- ***Budgets*** for debates, audits, and computation, including small models and few tool calls, as under limited connectivity;
- ***Answer types:*** from yes or no to score bands.

Regimes where the propositions say the methods fail are included, so failures are measured, not assumed; transfer is measured on held-out topics and model families.

Metrics, baselines, and statistics. Primary metrics are decision accuracy and calibration (Brier score and expected calibration error), wrong decisions under attack, correct minorities recovered and correct majorities overturned, audits needed to flag colluders, and cost in computation and tool calls. Each term of (1) is measured separately, so a gain can be traced to the task that produced it. Baselines, at matched compute, are a single AI judge (with and without the debaters' tools), majority vote, and averaging with equal weights; ablations remove each component in turn. Testbed B also reports the running example's measure: missed and false departure calls per 100 vessel questions, at each audit budget. We report means with confidence intervals over seeds and question samples.

Results and new knowledge. If successful, the project will produce:

(i) a reliability model that counts a group sharing a mistake at most once, with conditions for

Goal and benchmark	Success criterion
G1 (Task 1.1). Accuracy, calibration, and recovery of seeded groups across levels of sharing (A, B)	At least 5 points over majority vote when a third of agents share a source, growing with sharing, where three or more groups make mistakes independently (H1, Year 2)
G2 (Task 1.2). Gain in strategic raters' and coalitions' chance of being chosen; price of impartiality (PeerEval, C)	Coalition gains below a preset bound; impartiality costs less than debate adds (H2, Year 2)
G3 (Task 2.1). Accuracy on disputed questions; judge's error on each kind of step (B, C)	More accurate than open discussion and all baselines where a tool can settle the disputed steps (H3, Year 2)
G4 (Task 2.2). Wrong decisions that get through, time to flag, false flags, at red-team levels 1–4	Colluders cheating at a fixed rate flagged after a number of audits that does not grow with the stream; false flags below the chosen level (H4, Year 3)
G5 (Task 3). Held-out accuracy of decisions and first answers; error rate of the labels; share of the gain from training on known answers; rates of dishonest behavior	Larger gains than agreement and debate-win rewards, and at least half the gain from training on known answers, with no rise in dishonesty (H5, Year 3)
G6. Code, testbeds, protocols; peer-reviewed papers with proofs	Public release by month 36

Table 4: Goals, benchmarks (testbeds A–C, PeerEval), and success criteria, fixed before the experiments.

learning it without an answer key and rates for detecting groups;

(ii) an impartial mechanism for choosing debaters and judges, with its measured price and bounds on coalition gains;

(iii) an open agentic debate harness that reduces judging a question to judging single steps;

(iv) an audit scheme with bounds on the harm a colluder can do before it is flagged;

(v) a self-improvement method whose expected update follows accuracy rather than agreement.

Scientifically, it will show when a debate among AI agents can be trusted and how few audits keep the agents honest, which would let conferences and funders try impartial, audited peer review by AI; because people audit only a small share of decisions and training needs no labels, the methods can scale to AI whose answers people cannot easily evaluate.

Limitations. Where no tool applies, the judge rules on argument alone, and re-running verifies reported evidence, not its source. A misconception every agent shares cannot be detected from answers alone (Proposition 2), and training would reinforce it on a topic until that topic's human audits reveal it (Task 3). A group that cheats once is caught at most with the audit probability (Task 2.2). Paraphrasing limits but does not remove hidden signals (Mathew et al., 2025). Navy use would need Navy data and expert validation.

3.6 Project Schedule, Milestones, Goals, and Benchmarks

The project runs 36 months from April 1, 2027, with milestones M1–M6 (Table 3). Each goal has a benchmark and a criterion for success (Table 4); misses will be reported as findings.

3.7 Gold Standard Science

The project will follow the tenets of Executive Order 14303, as the PI's previous work does: MEQA, DSBench, and LMR-Bench were released with their data and code (Li et al., 2024b; Jing et al.,

2025; Yan et al., 2025), and PeerEval's leaderboard is public.

Reproducible and transparent. The harness logs every debate with exact tool calls, model versions, prompts, and seeds, and tool evidence is re-runnable by design. Code, testbeds, prompts, and protocols are released with each paper.

Communicative of error and uncertainty. Every guarantee is stated with its assumptions, decisions carry calibrated probabilities, and the framework's accuracy is estimated from random human audits with valid confidence intervals (Angelopoulos et al., 2023).

Skeptical and falsifiable. A red team attacks our own methods, ablations remove each component, and each task states a hypothesis that experiments can refute (H1–H5, Sections 3.2–3.4) and the regime in which its method should fail. Metrics and analysis plans will be fixed in advance.

Collaborative, peer reviewed, and open to negative results. The work joins statistics, social choice, AI safety, and natural language processing, with results for peer-reviewed venues in each; failures of debate or audits will be published as findings.

Without conflicts of interest. The PI discloses outside activities and support per UT Dallas policy.

4 Management Approach

This is a single-PI award with no subawardees, co-investigators, or senior personnel. The PI directs all research and is responsible for schedules, reports, and releases. The budget supports two Ph.D. students for three years, one month of PI summer effort per year, travel to ONR Project and Program Review Meetings and to conferences, and cloud computing and model access.

Team and roles. The project needs two kinds of expertise, so each student owns one. Student 1 (statistics and mechanism design) leads the models and their proofs: Tasks 1.1 and 1.2, the audit analysis of Task 2.2, and testbeds A and C. Student 2 (LLM agents and reinforcement learning) leads the systems and training: the harness and trained judges of Task 2.1, the planted-case generator of Task 2.2, Task 3, and testbed B, which reuses the MEQA pipeline. Both students run the red team, rotating roles so that no one attacks only their own method.

Coordination. The PI meets each student weekly and holds a weekly group meeting where the two lines of work connect: the reliability model and impartial choice feed the harness, and the harness's logged debates feed the audits and training, all in one shared code base. Every six months the team reviews progress against the milestones (Table 3) and success criteria (Table 4); a missed criterion triggers the fallback named in that task's evaluation.

Collaborators. There are no funded collaborators. The PI will seek feedback from Navy scientists through the placements described in Section 1.4.

Schedule. The project runs 36 months from April 1, 2027 (Table 3). Year 1 builds testbeds A and B, the harness, and the reliability model (M1, M2); Year 2 adds impartial choice, tool-backed debate, audits, and testbed C (M3, M4); Year 3 completes self-improvement and evaluates the full framework against the strongest red team (M5, M6).

Reporting and dissemination. The PI will submit annual and final reports, attend ONR Program Reviews, publish in peer-reviewed venues, and release code, testbeds, and protocols at M1, M6, and with each paper.

5 Principal Investigator Qualifications

Position and training. The PI, Xinya Du, has been an Assistant Professor of Computer Science at The University of Texas at Dallas since 2022. The PI received a Ph.D. in Computer Science from Cornell University in 2021, advised by Claire Cardie, and was a postdoctoral researcher with Heng Ji at the University of Illinois Urbana-Champaign. There the PI led the integration of RESIN, the event-prediction system built under a $10M DARPA KAIROS grant and the top-ranked system in the program's evaluation, and was first author of RESIN-11 (Du et al., 2022).

Recognition and service. NSF CAREER Award (2024–2029, funded on first submission); Amazon Research Award (2023); Cisco Research Award (2024); University of Chicago Rising Star in Data Science (2021). The PI's Ph.D. paper on neural question generation (Du et al., 2017) has been cited more than 1,000 times and is ranked by Paper Digest among the most influential ACL 2017 papers. The PI is an area chair for ICML 2027, ICLR 2027, and NeurIPS 2026, has been one for ICLR 2025, ACL, EMNLP (Outstanding Area Chair Award, 2024), and COLING, has served on NSF review panels, and co-organized the AI4Research workshop at AAAI 2025 and a multimodal knowledge workshop at IJCAI 2025.

Independent contributions since the Ph.D. With students at UT Dallas, the PI has built a line of work on AI evaluating AI (NeurIPS, ICLR, ACL, EMNLP, TMLR) that this project extends.

- *Peer evaluation among AI judges.* PRD (Li et al., 2024a) introduced peer ranking and peer discussion among AI judges, which agree with human judgments better than the strongest single judge and reduce a judge's bias toward its own model; researchers at NYU, Anthropic, and Cohere have built on it. It underlies PeerEval (peereval-ai.github.io), a public leaderboard on which seven AI models answer the same questions and judge one another anonymously (40,156 pairwise judgments in its first release). IQA-EVAL (Li et al., 2024c) evaluates interactive question answering with persona-driven AI agents.
- *Rewards and verification.* FG-PRM (Li et al., 2025) detects six types of hallucination at each step of a math solution; HiPRAG (Wu et al., 2026) rewards each search decision of an agent, cutting unnecessary searches to 2.3% while missing fewer needed ones; and Liang et al. (2026) show that a reasoning model can verify answers without further training.
- *Benchmarks and AI for science.* MEQA (Li et al., 2024b) provides event-centric multi-hop questions with explanations, FaithScore (Jing et al., 2024) measures fine-grained hallucinations, LMR-Bench (Yan et al., 2025) evaluates agents that reproduce research code, and DSBench (Jing et al., 2025), which OpenAI used to evaluate ChatGPT agent (OpenAI, 2025), tests agents on realistic data-science tasks; the PI co-wrote a survey of LLMs for scientific research, including peer review (Luo et al., 2025), and the group's research-idea generator appears at EMNLP 2026 (Li et al., 2026b).
- *Adversarial evaluation.* The PI co-advised one of the five university red teams in the Amazon Nova AI Challenge (Task 2.2) and holds an Open Philanthropy grant on control evaluations of AI debate.

6 Responsibility

Resources. UT Dallas provides the PI's laboratory space, a GPU server (ten A100 GPUs), and the shared Juno cluster (67 GPUs; see Facilities and Other Resources). With the requested cloud and model-access costs, these suffice; no equipment purchase is requested.

Ability to comply. The UT Dallas Office of Research and Innovation administers the PI's federal awards and will administer this award under 2 CFR 200 and ONR terms; the PI's commitments, listed in Current and Pending Support, leave the proposed effort available.

Performance history. The PI has managed an NSF CAREER award (2024–2029) and industry and foundation awards, filing reports on time; funds on current awards are being spent on schedule.

Integrity and business ethics. The PI and UT Dallas follow federal research-integrity policies (DoDI 3210.7); the PI has no record of misconduct.

Qualifications and eligibility. The PI is a U.S. permanent resident, has held a full-time tenure-track appointment at UT Dallas since 2022, the PI's first, and received the Ph.D. from Cornell University in 2021, on or after January 1, 2019, as Section B of the NOFO requires.

Organization and controls. UT Dallas maintains federally reviewed accounting, property-control, research-security, quality-assurance, and safety systems and is registered in SAM (UEI EJCVPNN1WFS5; CAGE 0W921); the research is computational.

Xinya Du

Curriculum Vitae · October 2026

Contact

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Research Impact

- **Citations.** 4,735 citations, h-index 29, i10-index 41 (Google Scholar, October 2026). "Learning to Ask" (ACL 2017) [51] has over 1,000 citations, is among Paper Digest's most influential ACL papers and the top 0.1% most-cited computer science articles of 2017, and was featured in *New Scientist* and *TechRepublic*; "Event Extraction by Answering (Almost) Natural Questions" (EMNLP 2020) [44] has over 600. Six papers from 2024–2025 already have over 150 citations each [1], [3], [5], [19], [21], [23].
- **Adoption beyond the lab.** OpenAI used DSBench [19] to evaluate ChatGPT agent (2025). PRD's peer evaluation of AI judges [5] has been built on by researchers at NYU, Anthropic, and Cohere; FaithScore [23] has been adopted by groups at Arizona State, the University of Washington, Johns Hopkins, and Intel Labs. Led the LLM usage-and-licensing section of *The Practical Guides for LLMs* on GitHub (10k+ stars).
- **Publications.** 66 papers and preprints: 5 journal articles (ACM Computing Surveys, IJCV, TMLR) and 46 peer-reviewed conference papers, 28 of them in the main proceedings of NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, EACL, and AAAI.
- **Recognition.** NSF CAREER Award (2024, funded on first submission); Best Paper Award, AAAI 2025 AI4Research Workshop; Best Poster Award (top 1%), ICML 2024 AI for Science Workshop; Outstanding Area Chair Award, EMNLP 2024; AAAI New Faculty Highlights (2024); Amazon (2023) and Cisco (2024) Research Awards; University of Chicago Rising Star in Data Science (2021).
- **Systems and resources.** First author of RESIN-11 [36]; led the integration of RESIN, the event-prediction system built under a $10M DARPA KAIROS grant and the top-ranked system in the program's evaluation. Built the PeerEval public leaderboard on PRD [5], and benchmarks including MEQA, DSBench, LMR-Bench, and SciEvent.
- **Community leadership.** Area Chair for ICML (2027), NeurIPS (2026), ICLR (2025, 2027), ACL, EMNLP, and COLING; NSF review panels, including CAREER (2025, 2026); co-organizer of tutorials at AAAI 2026 and ICMR 2025 and of workshops at AAAI 2025, CVPR 2025, and IJCAI 2025; over 20 invited talks and panels.
- **Funding.** $1.02M in external research funding as sole PI (NSF CAREER, Open Philanthropy, Amazon, Cisco, Adobe, OpenAI), plus Amazon awards as co-PI and lead contributor.
- **Mentoring and teaching.** Three Ph.D. students (two finishing in 2026) and over 20 master's, undergraduate, and K–12 students; former mentees went on to Ph.D. programs at CMU, Cornell, and NTU. An undergraduate mentee published in TMLR and won a Jonsson School award, and K–12 outreach projects led to the paper underlying the NSF CAREER award. New LLM course rated 4.75/5 (instructor) and 4.65/5 (course); President's Teaching Excellence Award nominee.
- **Most related to this proposal** ($\star$ in the publication list). AI judges that evaluate one another (PRD [5], IQA-EVAL [24]); step-level rewards and verification (FG-PRM [15], HiPRAG [10], training-free verification [12]); evaluation benchmarks (MEQA [22], FaithScore [23], LMR-Bench [14]); red-teaming of code models (COMET [57]); and AI for science (LDC [6], LLM4SR [1]).

Research Interests

Natural language processing and large language models: reasoning, post-training, and alignment; trustworthy evaluation and oversight of AI agents (peer evaluation, debate, process reward models); information extraction; AI for science.

Education

Cornell University Aug 2016 – Aug 2021
Ph.D. in Computer Science (M.S. degree granted in Aug 2019)
Dissertation: *Towards More Intelligent Extraction of Information from Documents*
Advisor: Claire Cardie. Worked closely with Sasha Rush.

Shanghai Jiao Tong University Sep 2012 – Aug 2016
B.E. in Computer Science and Engineering
Outstanding Graduates Award

Professional Experience

The University of Texas at Dallas Aug 2022 – Present
Assistant Professor (tenure-track), Department of Computer Science

University of Illinois at Urbana-Champaign Sep 2021 – Aug 2022
Postdoctoral Research Associate. Worked with Heng Ji.

Industrial Experience

Google AI Research May 2020 – Aug 2020
Research Intern. Supervisor: Luheng He, Qi Li, Emily Pitler.

Allen Institute for Artificial Intelligence (AI2) Sep 2018 – Dec 2018
Research Intern. Supervisor: Bhavana Dalvi, Peter Clark.

Microsoft Research (MSR) May 2018 – Aug 2018
Research Intern. Supervisor: Paul Bennett, Ahmed Awadallah.

Awards & Honors

Best Paper Award (for LDC [6]) AAAI AI4Research Workshop, 2025
Faculty Research Award Department of Computer Science, UT Dallas, 2025
NSF CAREER Award National Science Foundation, 2024
Outstanding Area Chair Award EMNLP, 2024
Best Poster Award (top 1% of 250 submissions; [21]) ICML AI for Science Workshop, 2024
AAAI New Faculty Highlights (paper [27]) AAAI, 2024
Cisco Faculty Research Award Cisco, 2024
President's Teaching Excellence Award Nominee UT Dallas, 2024
Amazon Research Award Amazon, 2023
Spotlight Rising Star in Data Science University of Chicago, 2021
Top 100 New Stars in Artificial Intelligence Baidu Scholar, 2020
Most Influential ACL Papers (15 each year; [51]) Paper Digest, 2017
Outstanding Graduates Award Shanghai Jiao Tong University, 2016
National Scholarship (top 1% of students nationwide) SJTU, 2013
National Olympiad in Mathematics in Provinces, First Prize 2011
National Olympiad in Informatics in Provinces, First Prize 2010

Paper-level recognition (oral presentations, top-1% and top-0.1% most-cited designations, media coverage, and the DARPA KAIROS result) is noted with each publication below.

Grants

Control Evaluations for AI Debate (Single PI)
Funding Source: Open Philanthropy.
Period: Oct 2025 – present.
Amount Awarded: $218,513 (+ $40,000 computing). Amount to me: $218,513.

NSF CAREER: Learning to Extract Consistent Event Graphs from Long and Complex Documents (Single PI)
Funding Source: National Science Foundation (NSF).
Period: May 2024 – present.
Amount Awarded: $561,219. Amount to me: $561,219.
Awarded on first submission.

Adobe Research Gift: Hallucinations in Video Language Models (Single PI)
Funding Source: Adobe.
Period: Sep 2025 – present.
Amount Awarded: $30,000.

Process-guided Fine-tuning for Answering Complex Questions (Single PI)
Funding Source: Amazon Research Award.
Period: Jan 2024 – present.
Amount Awarded: $110,000 ($40,000 AWS credits). Amount to me: $110,000.

Amazon Trusted AI Challenge Grant (Co-PI, with Prof. Wei Yang)
Funding Source: Amazon.
Period: Jan 2025 – present.
Amount Awarded to Our Team: $250,000.

Optimizing RISC-V Compilers with RISC-LLM and Syntax Parsing (Lead Contributor; PI: Wei Yang)
Funding Source: Amazon Research Award.
Period: Sep 2025 – Aug 2026.
Amount Awarded: $70,000. Amount to me: $23,000.

FAIGen: Faithful LLM Generation with Scientific Principles-guided Learning (Single PI)
Funding Source: Cisco Faculty Research Award.
Period: Jan 2025 – present.
Amount Awarded: $90,000. Amount to me: $90,000.

Mitigating Hallucinations in Vision-language Models with Fine-grained AI Feedback (Single PI)
Funding Source: OpenAI Researcher Access Program.
Period: Sep 2024 – present.
Amount Awarded: $12,000. Amount to me: $12,000.

Undergraduate Research Apprenticeship Award (URAP) (Single PI)
Funding Source: University of Texas at Dallas.
Period: Summer 2023, 2024, 2025.
Amount Awarded: $13,500. Amount to me: $13,500.

New Faculty Research Symposium Grant (Leading PI)
Funding Source: University of Texas at Dallas.
Period: Aug 2025 – Jul 2026.
Amount Awarded: $25,000.

Grants (Pending)

Collaborative Research: RI: SMALL: Explainable Methods for Natural Language Reasoning (Leading PI, co-PI: Prof. Claire Cardie)
Funding Source: National Science Foundation (NSF).
Period: Jan 2025 – Dec 2028.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending (**rated highly competitive on first submission**).

ACED: Large Language Model-based Research Agent for Synthetic Biology Research Idea Generation (Leading PI, co-PI: Prof. Leo Bleris)
Funding Source: National Science Foundation (NSF).
Period: Aug 2026 – Dec 2028.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending (**rated competitive on first submission**).

III: Automating Scientific Knowledge Discovery through Retrieval, Generation, and Validation (Leading PI, co-PI: Prof. Vincent Ng)
Funding Source: National Science Foundation (NSF).
Period: Jan 2027 – Dec 2029.
Amount Requested: $600,000. Amount to me: $300,000.
Status: Pending.

LLM-Guided Agents for Rational and Reliable Cell Reprogramming (Co-PI, co-PI: Prof. Leo Bleris)
Funding Source: National Science Foundation (NSF) Directorate for Biological Sciences (BIO).
Period: Aug 2026 – Dec 2029.
Amount Requested: $726,459. Amount to me: $242,153.
Status: Pending.

Publications

Notes: † co-leading author; * equal contribution; ⋆ work most related to the proposed ONR research. Current lists are also on [DBLP] and [Google Scholar]. In computer science, peer-reviewed conference proceedings are the primary archival venue for research; the main tracks of NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, EACL, and AAAI are highly selective, most accepting roughly 20–30% of submissions.

Journal Articles

[1] ⋆**LLM4SR: A Survey on Large Language Models for Scientific Research** [pdf]
Ziming Luo, Zonglin Yang, Zexin Xu, Wei Yang, **Xinya Du**
In ACM Computing Surveys (CSUR), to appear.

[2] **Dr.V: A Hierarchical Perception-Temporal-Cognition Framework to Diagnose Video Hallucination by Fine-Grained Spatial-Temporal Grounding** [pdf]
Meng Luo, Shengqiong Wu, Liqiang Jing, Tianjie Ju, Li Zheng, Jinxiang Lai, Tianlong Wu, **Xinya Du**, Jian Li, Siyuan Yan, Jiebo Luo, William Yang Wang, Hao Fei, Mong-Li Lee, Wynne Hsu
In International Journal of Computer Vision (IJCV), 2026.

[3] **SFT or RL? An Early Investigation into Training R1-Like Reasoning Large Vision-Language Models** [pdf]
Hardy Chen, Haoqin Tu, Fali Wang, Hui Liu, Xianfeng Tang, **Xinya Du**, Yuyin Zhou, Cihang Xie
In Transactions on Machine Learning Research (TMLR), 2025.

[4] **FGAIF: Aligning Large Vision-Language Models with Fine-grained AI Feedback** [pdf]
Liqiang Jing, **Xinya Du**
In Transactions on Machine Learning Research (TMLR), 2025.

[5] ⋆**PRD: Peer Rank and Discussion Improve Large Language Model-based Evaluations** [pdf]
Ruosen Li, Teerth Patel, **Xinya Du**
In Transactions on Machine Learning Research (TMLR), 2024.
Public leaderboard built on this work: peereval-ai.github.io

Peer-Reviewed Conference Papers (main, Findings, and demonstration tracks)

[6] ⋆**LDC: Learning to Generate Research Idea with Dynamic Control** [pdf]
Ruochen Li, Liqiang Jing, Chi Han, Jiawei Zhou, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.
Earlier version in AAAI 2025 AI4Research Workshop. **(Oral)**
Best Paper Award, AAAI 2025 AI4Research Workshop.

[7] **A Comprehensive Analysis for Visual Object Hallucination in Large Vision-Language Models** [pdf]
Liqiang Jing, Hardy Chen, Ehsan Aghazadeh, Xin Eric Wang, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2026.

[8] **A Skill-augmented Agentic Framework and Benchmark for Multi-Video Understanding** [pdf]
Yue Zhang, Liqiang Jing, Jia Li, Yapeng Tian, **Xinya Du**, Yunhui Guo, Vibhav Giridhar Gogate
In Findings of the Association for Computational Linguistics: EMNLP, 2026.

[9] **AMARIS: A Memory-Augmented Rubric Improvement System for Rubric-Based Reinforcement Learning** [pdf]
Peilin Wu, Xinlu Zhang, Kun Wan, Wentian Zhao, Gang Wu, **Xinya Du**, Zhiyu Chen
In Findings of the Association for Computational Linguistics: EMNLP, 2026.

[10] ⋆**HiPRAG: Hierarchical Process Rewards for Efficient Agentic Retrieval Augmented Generation** [pdf]
Peilin Wu, Mian Zhang, Kun Wan, Wentian Zhao, Kaiyu He, **Xinya Du**, Zhiyu Chen
In International Conference on Learning Representations (ICLR), 2026.

[11] **Is Grokking Worthwhile? Functional Analysis and Transferability of Generalization Circuits in Transformers** [pdf]
Kaiyu He, Mian Zhang, Peilin Wu, **Xinya Du**, Zhiyu Chen
In Findings of the Association for Computational Linguistics: ACL, 2026.

[12] ⋆**Your Reasoning Model is Secretly a Reward Model – Optimization-Free Verification from Experience** [pdf]
Zhenwen Liang, Ruosen Li, Yujun Zhou, Linfeng Song, Dian Yu, **Xinya Du**, Haitao Mi, Dong Yu
In Annual Meeting of the Association for Computational Linguistics (ACL), 2026.

[13] **FIFA: Unified Faithfulness Evaluation Framework for Text-to-Video and Video-to-Text Generation** [pdf]
Liqiang Jing, Viet Dac Lai, Seunghyun Yoon, Trung Bui, **Xinya Du**
In Findings of the Association for Computational Linguistics: ACL, 2026.

[14] ⋆**LMR-Bench: Evaluating LLM Agent's Ability on Reproducing Language Modeling Research** [pdf]
Shuo Yan, Ziming Luo, Zimu Wang, Ruochen Li, Daoyang Li, Liqiang Jing, Kaiyu He, Peilin Wu, Juntong Ni, George Michalopoulos, Yue Zhang, Ziyang Zhang, Mian Zhang, Zhiyu Chen, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[15] ⋆**FG-PRM: Fine-grained Hallucination Detection and Mitigation in Language Model Mathematical Reasoning** [pdf]
Ruosen Li, Ziming Luo, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2025.

[16] **SciEvent: Benchmarking Multi-domain Scientific Event Extraction** [pdf]
Bofu Dong, Pritesh Shah, Sumedh Sonawane, Tiyasha Banerjee, Erin Brady, **Xinya Du**, Ming Jiang
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[17] **Search Wisely: Mitigating Sub-optimal Agentic Searches By Reducing Uncertainty** [pdf]
Peilin Wu, Mian Zhang, Xinlu Zhang, **Xinya Du**, Zhiyu Chen
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025.

[18] **VistaDPO: Video Hierarchical Spatial-Temporal Direct Preference Optimization for Large Video Models** [pdf]
Haojian Huang*, Haodong Chen*, Shengqiong Wu, Meng Luo, Jinlan Fu, **Xinya Du**, Hanwang Zhang, Hao Fei
In International Conference on Machine Learning (ICML), 2025.

[19] **DSBench: How Far Are Data Science Agents from Becoming Data Science Experts?** [pdf]
Liqiang Jing, Zhehui Huang, Xiaoyang Wang, Wenlin Yao, Wenhao Yu, Kaixin Ma, Hongming Zhang, **Xinya Du**, Dong Yu
In International Conference on Learning Representations (ICLR), 2025.
Used by OpenAI to evaluate ChatGPT agent. [Link]

[20] **FIHA: Autonomous Hallucination Evaluation in Vision-Language Models with Davidson Scene Graphs** [pdf]
Bowen Yan, Zhengsong Zhang, Liqiang Jing, Eftekhar Hossain and **Xinya Du**
In Findings of the Association for Computational Linguistics: ACL, 2025.

[21] **Large Language Models for Automated Open-domain Scientific Hypotheses Discovery** [pdf]
Zonglin Yang, **Xinya Du**†, Junxian Li, Jie Zheng, Soujanya Poria, Erik Cambria
In Findings of the Association for Computational Linguistics: ACL, 2024.
Best poster award in ICML AI4Science, 2024 (Top 1% of 250 submissions).

[22] ⋆**Multi-hop Event-centric Question Answering with Explanations**
Ruosen Li, Zimu Wang, Son Quoc Tran, Lei Xia, **Xinya Du**
In Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

[23] ⋆**FaithScore: Fine-grained Evaluations of Hallucinations in Large Vision-Language Models** [pdf]
Liqiang Jing, Ruosen Li, Yunmo Chen, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2024.

[24] ⋆**IQA-EVAL: Automatic Evaluation of Human-Model Interactive Question Answering** [pdf]
Ruosen Li, Ruochen Li, Barry Wang, **Xinya Du**
In Annual Conference on Neural Information Processing Systems (NeurIPS), 2024.

[25] **QAEvent: Event Extraction as Question-Answer Pairs Generation** [pdf]
Milind Choudhary, **Xinya Du**
In Findings of the Association for Computational Linguistics: EACL, 2024.

[26] **Language Models as Inductive Reasoners** [pdf]
Zonglin Yang, Li Dong, **Xinya Du**†, Hao Cheng, Erik Cambria, Xiaodong Liu, Jianfeng Gao, Furu Wei
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2024. **(Oral)**

[27] **Making Natural Language Reasoning Explainable and Faithful** [pdf]
Xinya Du
In AAAI Conference on Artificial Intelligence (AAAI), 2024. New Faculty Highlights. **(Oral)**

[28] **Proto-CLIP: Vision-Language Prototypical Network for Few-Shot Learning** [pdf]
Jishnu Jaykumar P, Kamalesh Palanisamy, Yu-Wei Chao, **Xinya Du**, Yu Xiang
In International Conference on Intelligent Robots and Systems (IROS), 2024. **(Oral)**

[29] **Document-level Causal Relation Extraction with Knowledge-guided Binary Question Answering** [pdf]
Zimu Wang, Lei Xia, Wei Wang, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2024.

[30] **Leveraging Structured Information for Explainable Multi-hop Question Answering and Reasoning** [pdf]
Ruosen Li, **Xinya Du**
In Findings of the Association for Computational Linguistics: EMNLP, 2023.

[31] **Process of Elimination for Multiple Choice Reasoning** [pdf]
Chenkai Ma, **Xinya Du**
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.

[32] **Probing Representations for Document-level Event Extraction** [pdf]
Barry Wang, **Xinya Du**, Claire Cardie
In Findings of the Association for Computational Linguistics: EMNLP, 2023.

[33] **End-to-end Case-Based Reasoning for Commonsense Knowledge Base Completion** [pdf]
Zonglin Yang, **Xinya Du**†, Erik Cambria, Claire Cardie
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2023. **(Oral)**

[34] **Zero-Shot Classification by Logical Reasoning on Natural Language Explanations** [pdf]
Chi Han, Hengzhi Pei, **Xinya Du**, Heng Ji
In Findings of the Association for Computational Linguistics: ACL, 2023.

[35] **Logical Entity Representation in Knowledge-Graphs for Differentiable Rule Learning** [pdf]
Chi Han, Qizheng He, Charles Yu, **Xinya Du**, Hanghang Tong, Heng Ji
In International Conference on Learning Representations (ICLR), 2023.

[36] **RESIN-11: Schema-guided Event Prediction for 11 Newsworthy Scenarios** [pdf]
Xinya Du, Zixuan Zhang, Sha Li, Heng Ji, the RESIN team
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): System Demonstrations, 2022.
Top ranking system in DARPA KAIROS evaluation. [Link]

[37] **Retrieval-Augmented Generative Question Answering for Event Argument Extraction** [pdf]
Xinya Du, Heng Ji
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2022.

[38] **Dynamic Global Memory for Document-level Argument Extraction** [pdf]
Xinya Du, Sha Li, Heng Ji
In Annual Meeting of the Association for Computational Linguistics (ACL), 2022.

[39] **Automatic Error Analysis for Document-level Information Extraction** [pdf]
Aliva Das*, **Xinya Du***, Barry Wang*, Kejian Shi, Jiayuan Gu, Thomas Porter, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2022.

[40] **Template Filling with Generative Transformers** [pdf]
Xinya Du, Alexander M. Rush, Claire Cardie
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021.

[41] **GRIT: Generative Role-filler Transformers for Document-level Event Entity Extraction** [pdf]
Xinya Du, Alexander M. Rush, Claire Cardie
In Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2021. **(Oral)**

[42] **Few-shot Intent Classification and Slot Filling with Retrieved Examples** [pdf]
Dian Yu, Luheng He, Yuan Zhang, **Xinya Du**, Panupong Pasupat, Qi Li
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2021. **(Oral)**

[43] **QA-Driven Zero-shot Slot Filling with Weak Supervision Pretraining** [pdf]
Xinya Du, Luheng He, Qi Li, Dian Yu, Panupong Pasupat, Yuan Zhang
In Annual Meeting of the Association for Computational Linguistics (ACL), 2021.

[44] **Event Extraction by Answering (Almost) Natural Questions** [pdf]
Xinya Du, Claire Cardie
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020. **(Oral)**
Top 1% most cited articles published in Computer Science in 2020.

[45] **Improving Event Duration Prediction via Time-aware Pre-training** [pdf]
Zonglin Yang, **Xinya Du**, Alexander M. Rush, Claire Cardie
In Findings of the Association for Computational Linguistics: EMNLP, 2020.

[46] **Document-Level Event Role Filler Extraction using Multi-Granularity Contextualized Encoding** [pdf]
Xinya Du, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2020.

[47] **Leveraging Structured Metadata for Improving Question Answering on the Web** [pdf]
Xinya Du, Adam Fourney, Robert Sim, Claire Cardie, Paul Bennett, Ahmed Hassan Awadallah
In Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (AACL/IJCNLP), 2020.

[48] **Be Consistent! Improving Procedural Text Comprehension using Label Consistency** [pdf]
Xinya Du, Bhavana Dalvi, Niket Tandon, Antoine Bosselut, Wen-tau Yih, Peter Clark, Claire Cardie
In Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2019.

[49] **Harvesting Paragraph-Level Question-Answer Pairs from Wikipedia** [pdf]
Xinya Du, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2018. **(Oral)**
Top 1% most cited articles published in Computer Science in 2018.

[50] **Identifying Where to Focus in Reading Comprehension for Neural Question Generation** [pdf]
Xinya Du, Claire Cardie
In Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017.

[51] **Learning to Ask: Neural Question Generation for Reading Comprehension** [pdf]
Xinya Du, Junru Shao, Claire Cardie
In Annual Meeting of the Association for Computational Linguistics (ACL), 2017.
Most Influential ACL Papers (15 each year). [Link]
Featured in New Scientist and TechRepublic. [Link]
Top 0.1% most cited articles published in Computer Science in 2017.

Workshop Papers, Tutorial Abstracts, and Shared-Task Reports

[52] **Efficient Document-level Event Relation Extraction**
Ruochen Li, Zimu Wang, **Xinya Du**
In 10th Workshop on Representation Learning for NLP at Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2025.

[53] **Hallucinations in Large Language Models and Large Vision-Language Models (Tutorial)** [pdf]
Liqiang Jing, Yue Zhang, **Xinya Du**
In Proceedings of the ACM International Conference on Multimedia Retrieval (ICMR), pp. 2138–2139, 2025.

[54] **Logical Reasoning over Natural Language as Knowledge Representation: A Survey** [pdf]
Zonglin Yang, **Xinya Du**, Rui Mao, Jinjie Ni, Erik Cambria
In Workshop on Natural Language Reasoning and Structured Explanations (NLRSE) at Annual Meeting of the Association for Computational Linguistics (ACL), 2023.

[55] **Toward Consistent and Informative Event-Event Temporal Relation Extraction** [pdf]
Xiaomeng Jin, Haoyang Wen, **Xinya Du**, Heng Ji
In MATCHING at Annual Meeting of the Association for Computational Linguistics (ACL), 2023. **(Oral)**

[56] **Cornell Belief and Sentiment System at TAC 2016** [pdf]
Vlad Niculae, Kai Sun, Xilun Chen, Yao Cheng, **Xinya Du**, Esin Durmus, Arzoo Katiyar, Claire Cardie
In Text Analysis Conference (TAC), 2016.

Challenge Proceedings

[57] ⋆**COMET: Closed-loop Orchestration for Malicious Elicitation Techniques in Code Models** [pdf]
Z. Xu, T. Li, R. S. Rathnasuriya, Z. Song, J. Ren, B. Mandalapu, S. Setayeshpour, **Xinya Du**, Wei Yang
In 1st Proceedings of Amazon Nova AI Challenge (Trusted AI), 2025.

Preprints

[58] **Be Consistent! Enhancing Robust Visual Reasoning in LVLMs with Consistency Constraints** [pdf]
Liqiang Jing, Xiong Zhou, Siddharth Varia, Neha Anna John, **Xinya Du**, Vassilis N. Ioannidis
arXiv:2607.21722, 2026.

[59] **Event Extraction in Large Language Model** [pdf]
Bobo Li, Xudong Han, Jiang Liu, Yuzhe Ding, Liqiang Jing, Zhaoqi Zhang, Jinheng Li, **Xinya Du**, Fei Li, Meishan Zhang, Min Zhang, Aixin Sun, Philip S. Yu, Hao Fei
arXiv:2512.19537, 2025.

[60] **GEAR: A General Evaluation Framework for Abductive Reasoning** [pdf]
Kaiyu He, Peilin Wu, Mian Zhang, Kun Wan, Wentian Zhao, **Xinya Du**, Zhiyu Chen
arXiv:2509.24096, 2025.

[61] **AALC: Large Language Model Efficient Reasoning via Adaptive Accuracy-Length Control** [pdf]
Ruosen Li, Ziming Luo, Quan Zhang, Ruochen Li, Ben Zhou, Ali Payani, **Xinya Du**
arXiv:2506.20160, 2025.

[62] **Multimodal Reference Visual Grounding** [pdf]
Yangxiao Lu, Ruosen Li, Liqiang Jing, Jikai Wang, **Xinya Du**, Yunhui Guo, Nicholas Ruozzi, Yu Xiang
arXiv:2504.02876, 2025.

[63] **Do Retrieval-Augmented Language Models Adapt to Varying User Needs?** [pdf]
Peilin Wu, Xinlu Zhang, Wenhao Yu, Xingyu Liu, **Xinya Du**, Zhiyu Zoey Chen
arXiv:2502.19779, 2025.

[64] **MLR-Copilot: Autonomous Machine Learning Research based on Large Language Models Agents** [pdf]
Ruochen Li, Teerth Patel, Qingyun Wang, **Xinya Du**
arXiv:2408.14033, 2024.
The first autonomous LLM agent for the entire machine learning research process: generating hypotheses, implementing ideas, and running experiments.

[65] **VIA: Unified Spatiotemporal Video Adaptation Framework for Global and Local Video Editing** [pdf]
Jing Gu, Yuwei Fang, Ivan Skorokhodov, Peter Wonka, **Xinya Du**, Sergey Tulyakov, Xin Eric Wang
arXiv:2406.12831, 2024.

[66] **AGent: A Novel Pipeline for Automatically Creating Unanswerable Questions** [pdf]
Son Quoc Tran, Gia-Huy Do, Phong Nguyen-Thuan Do, Matt Kretchmar, **Xinya Du**
arXiv:2309.05103, 2023.

Dissertation

[67] **Towards More Intelligent Extraction of Information from Documents**
Xinya Du
Ph.D. Dissertation, Cornell University, 2021.

Professional Services

Grant-Proposal Review Panels

National Science Foundation (NSF) review panel, IIS/RI CAREER program, 2026.

National Science Foundation (NSF) review panel, IIS/RI Future CORe program, 2026.

National Science Foundation (NSF) review panel, IIS/RI MEDIUM program, 2025.

Area Chair Roles

ICML (International Conference on Machine Learning) Area Chair, 2027.

NeurIPS (Conference on Neural Information Processing Systems) Area Chair, 2026.

ICLR (International Conference on Learning Representations) Area Chair, 2025, 2027.

ACL (Annual Meeting of the Association for Computational Linguistics) Area Chair, 2023, 2025; ACL Rolling Review Area Chair, 2024; ACL Demo Track Area Chair, 2026.

EMNLP (Conference on Empirical Methods in Natural Language Processing) Area Chair, 2024 **(Outstanding Area Chair Award)**, 2025; EMNLP Demo Track Area Chair, 2024.

COLING (International Conference on Computational Linguistics) Area Chair, LREC-COLING 2024 and COLING 2025.

Conference Organizing Roles

NAACL (Conference of the North American Chapter of the Association for Computational Linguistics) Website Chair, 2024.

ACL Session Chair, 2024.

Tutorials and Workshops Organized

Frontiers of LLMs for Automated Scientific Discovery: A Tutorial
Organizing team: Zonglin Yang, Chandan Reddy, Xinya Du and Mengdi Wang.
In AAAI Conference on Artificial Intelligence (AAAI), 2026.

Hallucinations in Large Language Models and Large Vision-Language Models: A Tutorial
Organizing team: Liqiang Jing, Yue Zhang, Xinya Du.
In ACM International Conference on Multimedia Retrieval (ICMR), 2025.

AI4Research: Towards a Unified Knowledge-grounded Scientific Research Lifecycle (workshop).
Organizing team: Qingyun Wang, Wenpeng Yin, Lifu Huang, Yi R. Fung, Xinya Du, Carl Edwards, Tom Hope.
In AAAI Conference on Artificial Intelligence (AAAI), 2025.

Evaluations and Benchmarks in Context of Multimodal LLM (workshop).
Organizing team: Hao Fei, Xiang Yue, Kaipeng Zhang, Long Chen, Jian Li, Xinya Du.
In The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

The First Workshop on Multimodal Knowledge and Language Modeling
Organizing team: Liqiang Jing, Xinya Du, Hao Fei, Jing Gu, Manling Li, Aixin Sun, William Wang
In International Joint Conference on Artificial Intelligence (IJCAI), 2025.

Program Committee Member / Reviewer

ACL (Annual Meeting of the Association for Computational Linguistics).

NeurIPS (Conference on Neural Information Processing Systems).

ICLR (International Conference on Learning Representations).

AISTATS (International Conference on Artificial Intelligence and Statistics).

EMNLP (Conference on Empirical Methods in Natural Language Processing).

NAACL (Conference of the North American Chapter of the Association for Computational Linguistics).

AACL/IJCNLP (Asia-Pacific Chapter of the Association for Computational Linguistics / International Joint Conference on Natural Language Processing).

IJCAI (International Joint Conference on Artificial Intelligence).

AAAI (AAAI Conference on Artificial Intelligence).

NLPCC (Natural Language Processing and Chinese Computing).

CoNLL (Conference on Computational Natural Language Learning).

WNUT (Workshop on Noisy User-generated Text).

MRQA (Workshop on Machine Reading for Question Answering).

SEM (Joint Conference on Lexical and Computational Semantics).

Journal Reviewer

TKDE (IEEE Transactions on Knowledge and Data Engineering).

TASLP (IEEE Transactions on Audio, Speech and Language Processing).

TNNLS (IEEE Transactions on Neural Networks and Learning Systems).

TALLIP (ACM Transactions on Asian and Low-Resource Language Information Processing).

TKDD (ACM Transactions on Knowledge Discovery from Data).

CL (Computational Linguistics).

KAIS (Knowledge and Information Systems).

AI Communication.

IPM (Information Processing and Management).

Ph.D./M.S. Thesis Committee Member

Ph.D.: Kangshuo Li, Souradeep Nanda, Ali Riahi, Xiangci Li, Jishnu Jaykumar Padalunkal, Basel Abdeen, Yibo Hu.

M.S.: Wooseong Yang, Shubham Patel.

Department, University, and Outreach Service

Faculty advisor and judge: Association for Computing Machinery (ACM) Symposium, UT Dallas, 2023–present.

Faculty Advisor: Women Who Compute (WWC), UT Dallas, 2024–present.

Faculty Advisor: Society of Asian Scientists and Engineers (SASE), UT Dallas, 2024–present.

Invited Speaker: UTD AI Society's "AI Chats", 2025.

Panelist: FuturED 2024 Workshop on Future of Event Detection, 2024.

Faculty Host: K-12 Outreach, UT Dallas, Summer 2023, 2024.

Research Mentor: Research, Inquiry, Design Experience (RIDE) Project, UT Dallas, 2024.

PhD Admission Committee Member, UT Dallas CS Department, 2022, 2023, 2024.

Site Host: North American Computational Linguistics Olympiad (NACLO), 2023, 2024.

PhD Admission Committee Member, Cornell CS Department, 2021.

Volunteer: Cornell CS Department PhD Visit Day, 2019, 2020, 2021.

Student Volunteer: ACL, 2017, 2018.

Student Volunteer: EMNLP, 2017.

Invited Talks and Panels

Pushing the Boundaries of LLM Agents in Data Science and Machine Learning Research.
Nvidia, Dec 2025.

Inside LLMs: Knowledge, Alignment, and Agents.
Samsung Research America (SRA), May 2025.

Towards Autonomous Scientific Discovery with Large Language Models.
UT Southwestern AI in Medicine Seminar Series, May 2025.

Synergizing Knowledge and Large Language Models.
University of Illinois Urbana-Champaign, Data Mining Group Seminar, Oct 2024.
University of Massachusetts–Amherst, Machine Learning & Friends Lunch, Sep 2024.

Synergy between Large Language Model and Knowledge.
University of North Texas, Oct 2024.

Panelist, Future of Event Detection Workshop.
Conference on Empirical Methods in Natural Language Processing (EMNLP), Nov 2024.

Faculty Round Table Talk.
UT Dallas Hobson Wildenthal Honors College, Aug 2024.

Large Language Models: Knowledge, Reasoning and Factuality.
Samsung Electronics America, Mar 2024.

Making Natural Language Reasoning Explainable and Faithful (New Faculty Highlights).
AAAI Conference on Artificial Intelligence (AAAI), Feb 2024.

Open-ended Evaluations of Foundational Models: Alignment and Faithfulness.
Shanghai Jiao Tong University, Computer Science Department Seminar, Dec 2023.
Fudan University, Computer Science Department Seminar, Dec 2023.

Panelist, "ChatGPT: Fact vs. Fiction" public forum, sponsored by The Dallas Morning News.
UT Dallas, Mar 2023.

Towards More Intelligent Extraction of Information from Documents.
Allen Institute for Artificial Intelligence, Mar 2022.
Facebook, Mar 2022.
Simon Fraser University, Feb 2022.
Rensselaer Polytechnic Institute, Feb 2022.
University of California, Merced, Feb 2022.
UIUC, Siebel School of Computing and Data Science Speaker Series, Feb 2022.

Towards More Informed Extraction of Events from Documents.
University of Chicago, Rising Stars in Data Science Workshop, Jan 2021.
Tencent AI Research America, Nov 2020.

Event Extraction by Answering (Almost) Natural Question.
UIUC, Information Extraction and Knowledge Acquisition Class, Sep 2020.

LwLL: Progress on the NLP Front.
Cornell University, DARPA site visit, Apr 2020.

Mentoring Experience

PhD students

Ruosen Li (2022–2026, UT Dallas PhD student, graduated Aug 2026).
Publications: ACL 2026; NeurIPS 2024 (two papers); TMLR 2024; EMNLP Findings 2023, 2024, 2025.

Liqiang Jing (2023–2026, UT Dallas PhD student, graduating Nov 2026).
Publications: ICLR 2025; EMNLP 2025, 2026; TMLR 2025; IJCV 2026; EMNLP Findings 2024, 2026 (two papers); ACL Findings 2025, 2026; ICMR 2025 tutorial.
Award: OpenAI Researcher Access Program.

Ruochen Li (2023–present, UT Dallas PhD student).
Publications: EMNLP 2026 (first author; an earlier version won the Best Paper Award at the AAAI 2025 AI4Research Workshop); EMNLP 2025; NeurIPS 2024; RepL4NLP Workshop at NAACL 2025.

Master Students

Milind Choudhary (2023, UT Dallas Master student) → UT Dallas PhD.
Topic: Event Extraction as Question-Answer Pairs Generation.
Publications: EACL Findings 2024.

Zimu Wang (2023–present, UT Dallas Visiting student) → UoL PhD.
Topic: Temporal relation extraction.
Publications: NeurIPS 2024, EMNLP 2025, EMNLP Findings 2024, RepL4NLP Workshop at NAACL 2025.

Son Tran (2023, UT Dallas Visiting student) → Cornell PhD.
Topic: Complex Question Answering.
Publications: NeurIPS 2024.

Chenkai Ma (2023, UT Dallas Visiting student).
Topic: Multiple Choice Questions Reasoning.
Publications: EMNLP 2023.

Barry Wang (2021–2024, Cornell CS undergraduate student) → CMU PhD.
Topic: Automatic Error Analysis for Information Extraction.
Publications: ACL 2022, SciNLP 2022, EMNLP Findings 2023, NeurIPS 2024.

Zonglin Yang (2020–2024, Cornell CS MEng student) → NTU PhD.
Topic: Commonsense and Case-based Reasoning for NLP.
Publications: EMNLP Findings 2020, EACL 2023, EACL 2024, ACL Findings 2024 (Best Poster Award, ICML 2024 AI for Science Workshop).

Ehsan Aghazadeh (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Interpretability.
Publications: EMNLP Findings 2026.

Bowen Yan (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Hallucinations.
Publications: ACL Findings 2025.

Zhengsong Zhang (2024, UT Dallas Visiting student).
Topic: Large Vision Language Models Hallucinations.
Publications: ACL Findings 2025.

Ziming Luo (2024, UT Dallas Visiting student).
Topic: Large Language Model for Scientific Discovery.
Publications: EMNLP 2025, EMNLP Findings 2025.

Undergraduate/High School Students

Teerth Patel (UT Dallas BS student, 2023–present).
Topic: Large Language Models Peer Evaluations.
Publications: TMLR 2024.
Award: Jonsson School of Engineering and Computer Science Award.

Lei Xia (2023–2024, UT Dallas Visiting student).
Topic: Multi-hop Question Answering.
Publications: NeurIPS 2024, EMNLP Findings 2024.

Minhao Zou (2024, UT Dallas Visiting student).
Topic: LLM for Scientific Discovery.

Arjun Junghare (2024, UT Dallas CS undergraduate student).
Topic: Image Generation with Large Language Models.

Shreya Kumara (2024, President of UTD Society of Asian Scientists and Engineers).

Anjana Bharadwaj (2024, President of UTD Women Who Compute).

Jaden Nunes (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Rishab Bhattacharya (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Shreyas Kumar (Summer 2023, DFW Local K-12 student).
Topic: Event Extraction as Question-Answer Pairs Generation.

Aliva Das (2021–2022, Cornell CS undergraduate student) → Amazon.
Topic: Automatic Error Analysis for Information Extraction.
Publications: ACL 2022, SciNLP 2021.

Rishi Malhotra (Spring 2021, Cornell CS undergraduate student) → Microsoft.
Topic: Applying Neural Document-level IE Model to the Scientific Domain.

Maitreyi Chatterjee (Spring 2021, Cornell CS undergraduate student) → LinkedIn.
Topic: Applying Neural Document-level IE Model to the Scientific Domain.

Teaching Experience

Introduction to Machine Learning, UT Dallas, Spring 2024.

Natural Language Processing, UT Dallas, Fall 2023, 2024, 2025.

Deep Learning for Natural Language Processing, UT Dallas, Spring 2023.
New course developed and taught (overall instructor score 4.75/5.0; course score 4.65/5.0). [Link]

Natural Language Processing, UT Dallas, Fall 2022.

Natural Language Processing, Cornell University, Fall 2019.
Teaching Assistant for Prof. Claire Cardie.

Natural Language Processing, Cornell University, Spring 2019.
Teaching Assistant for Prof. Yoav Artzi.

Software Engineering, Cornell University, Spring 2017, 2018.
Teaching Assistant for Prof. William Arms.

Introduction to Computing Using Python, Cornell University, Fall 2016.
Teaching Assistant for Prof. Walker White.

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COMPUTER SCIENCE DEPARTMENT

ERIK JONSSON SCHOOL OF
ENGINEERING AND COMPUTER SCIENCE

October 20, 2026

Dear ONR Young Investigator Program Selection Committee:

I am pleased to provide my strongest endorsement for Dr. Xinya Du's application to the Office of Naval Research Young Investigator Program, entitled "Trustworthy Collective Reasoning and Evaluation for AI Agents." As an Assistant Professor in the Department of Computer Science at The University of Texas at Dallas, Dr. Du is an outstanding early-career researcher with exceptional potential to become a leader in artificial intelligence, with a focus on language model reasoning, evaluation, and autonomous agents. Our department is strongly committed to supporting his research and long-term academic career.

Dr. Du was recruited to UT Dallas based on his strong research accomplishments, technical expertise, and ambitious vision for advancing artificial intelligence. His recruitment reflects our department's commitment to attracting outstanding researchers with the potential to make significant contributions to their fields. Since joining the department, he has established an active research program addressing fundamental challenges in the reliability and reasoning capabilities of modern AI systems. His research has resulted in publications at leading conferences in natural language processing and machine learning, including NeurIPS, ACL and EMNLP. His ability to identify important emerging problems and develop original technical solutions distinguishes him as a highly promising young investigator.

Dr. Du's proposed ONR research, "Trustworthy Collective Reasoning and Evaluation for AI Agents," addresses a fundamental challenge in the development of reliable autonomous AI systems: enabling multiple agents to evaluate one another's reasoning, recognize relevant expertise, and improve collective decisions even when individual agents are imperfect or unreliable. The project seeks to develop novel methods for peer evaluation, collective reasoning, and trustworthy collaboration among AI agents. This research has significant potential to advance the foundations of multi-agent intelligence and contribute to reliable autonomous decision-making in complex environments. Dr. Du's expertise in language model reasoning and evaluation, together with his ongoing work on AI agents, positions him well to pursue this ambitious research agenda.

The University of Texas at Dallas is committed to providing sustained institutional support for Dr. Du's research and career development. The department has provided him with start-up funding, dedicated research space, and access to computing resources. We also support his research through a reduced teaching workload, support for graduate student recruitment, faculty mentoring, and professional development opportunities. These institutional investments will help Dr. Du further develop his independent research program and expand its impact. Our commitment extends beyond the proposed ONR award, reflecting our confidence in his long-term potential and our investment in his development as a leading faculty member.

Dr. Du exemplifies the qualities sought by the ONR Young Investigator Program: scientific creativity, technical excellence, research independence, and exceptional promise for sustained contributions to his

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field. I strongly endorse his application and reaffirm our department's commitment to supporting his continued success at The University of Texas at Dallas.

Sincerely,

Rym Zalila-Wenkstern



Rym Zalila-Wenkstern, Ph.D.
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9 Data Management Plan

Types of data, software, and other materials. The project will produce (1) three testbeds of questions with known answers: math and code problems with planted shared mistakes (A), multi-document questions composed from event structures, including synthetic maritime scenarios (B), and scientific-claim and peer-review questions with planted errors (C); (2) logs of agent answers, ratings, debates, tool calls and their outputs, judge rulings, and audit outcomes; (3) software: the agentic debate harness, the shared-error-factor reliability model, the impartial selection mechanism, the audit scheme, red-team agents, reinforcement-learning code, and the simulation code behind the theoretical results; and (4) fine-tuned weights of open-weight models, where the original licenses permit release.

How the data will be acquired. Testbed A draws on public benchmarks (GSM8K and MATH, released under the MIT license) and public programming problems. Testbed B is generated by our MEQA pipeline from event structures; its maritime scenarios are fully synthetic and unclassified. Testbed C uses public papers and the public reviews and rebuttals posted on OpenReview, under the platform's terms, together with published reproducibility reports. Logs are produced by running agents built on open-weight models on UT Dallas computers and on commercial models through their interfaces. No human-subjects data are collected; in experiments, human audits are simulated by revealing known answers.

Time and location of acquisition. Not scientifically pertinent: all data are generated or collected computationally at UT Dallas during the 36-month project.

Processing. Questions are normalized to a common schema, deduplicated, and split into development, evaluation, and held-out sets by topic and by model family. Planted cases are generated by versioned scripts that record each edit and its effect on the answer. Logs are stored as produced and then summarized for analysis; every number in a publication is regenerated by a released script from the stored logs.

File formats and naming conventions. Questions and logs are stored as JSON Lines and Parquet; code is Python, released under the Mozilla Public License 2.0, as in the PI's earlier projects; model weights use the safetensors format. File names give the testbed, split, and version (for example, `tbB_eval_v1.2.jsonl`), with semantic versioning and a change log for each release.

Quality assurance and quality control. The known answer of every generated question is confirmed by construction and by an independent script; two students review a random sample of each release, and disagreements are resolved before release. Each log record carries the model version, prompt version, random seed, and a hash of tool outputs, so that any debate can be re-run. The harness is frozen at each milestone, and automated runs confirm that frozen versions reproduce stored results.

Origin of existing data. GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), the MEQA benchmark and pipeline (Li et al., 2024b), public OpenReview reviews, published reproducibility reports, and PeerEval logs produced by the PI's group. Each release records the source and license of every component.

Standards for data and metadata. Datasets will carry Croissant machine-learning metadata and a datasheet describing their purpose, composition, collection, and known limitations; code repositories will include environment files that pin software versions.

Preservation and access. Code will be public on GitHub, with each milestone release archived on Zenodo under a DOI. Testbeds and logs will be released on Hugging Face and on project webpages linked from the PI's lab website, by the time of the associated publication, and kept available for at least five years after the project ends. Working copies stay on access-controlled lab machines, with secondary copies on departmental systems at least once per semester; HPC scratch storage is never used as an archive. Final reports will be submitted as required by DoD Instruction 3200.12.

Balance of value and cost. We will keep all data behind published results. Raw logs of exploratory runs, which can be large, will be kept for one year and then reduced to summaries, since the released code can

regenerate them.

Security. All data are public or synthetic, unclassified, and contain no controlled unclassified information or personally identifiable information. If any data were later found to raise national security concerns, they would not be released, in accordance with DoD Instruction 5230.09.

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